Agent-Memory 评测全景：基准、评估与记忆系统（理论篇）

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本文系统梳理了Agent长期记忆能力的评测全景，涵盖基准数据集、评估框架与记忆系统三大核心维度。在基准方面，介绍了MUSE、LOCOMO等贴近真实交互的数据集；在评估方面，分析了MemoryAgentBench、LONGMEMEVAL及MemBench等框架，重点考察准确检索、长程理解、冲突解决及反思记忆等关键能力；在系统实现上对比了THEANINE、RMM、M3-Agent及Mem0等代表性方案的技术机制与性能表现。文章指出当前技术虽在检索准确性上有所进展，但在跨会话推理、动态更新及效率平衡上仍存瓶颈，并强调未来评测需统一口径，综合考量检索正确性、使用有效性、时间维度及成本约束，以真正指导工程落地。

01 引言

随着大语言模型（LLM）在对话系统与智能代理中的应用加深，长期记忆能力正成为影响真实效能的关键因素。尽管LLM擅长短上下文生成，但在多轮、跨会话甚至多模态交互中仍常出现遗忘、推理断裂与一致性缺失。如何构建、更新与检索长期记忆，使模型能持续保留关键信息并适应变化，已成为重要挑战。

近年研究从三条主线推进：一是提出更贴近真实交互的基准与数据集（如MUSE、LOCOMO），二是建立更系统的评估框架（如MemoryAgentBench、LONGMEMEVAL、MemBench），三是探索更有效的记忆方法与系统（如THEANINE、RMM、Mem0，以及面向多模态场景的M3-Agent）。这些工作共同揭示了当前技术在检索准确性、跨会话推理、冲突处理与效率上的优势与瓶颈。

本文将从基准数据集、方法框架与记忆系统三方面梳理代表性进展，比较其设计思路、记忆机制与实验结论，并讨论未来应如何在更真实场景、动态更新与结构化表示之间做出更优权衡。

02 技术概况

Memory Benchmark

	来源	发表（统计截止2026年2月）
MUSE	Northeastern University	ACL 2025，被引次数5
LOCOMO	University of North Carolina	ACL 2024，被引次数274

Memory Evaluation

	来源	发表（统计截止2026年2月）
MemoryAgentBench	UC San Diego	arxiv，被引次数43
LONGMEMEVAL	UCLA, Tencent	arxiv，被引次数141
MemBench	Huawei	ACL 2025，被引次数23

Memory System

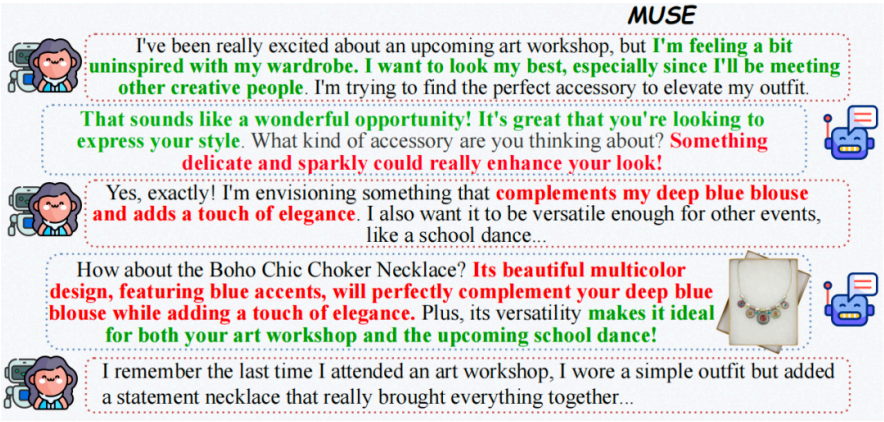
	来源	发表（统计截止2026年2月）
THEANINE&TeaFarm	Yonsei University	NAACL 2025，被引次数23
RMM	Google	ACL 2025，被引次数35
M3-Agent	ByteDance-Seed	ICLR 2026，被引次数29
Mem0	mem0ai	ECAI 2026，被引次数222

MUSE

《MUSE: A Multimodal Conversational Recommendation Dataset with Scenario-Grounded User Profiles》

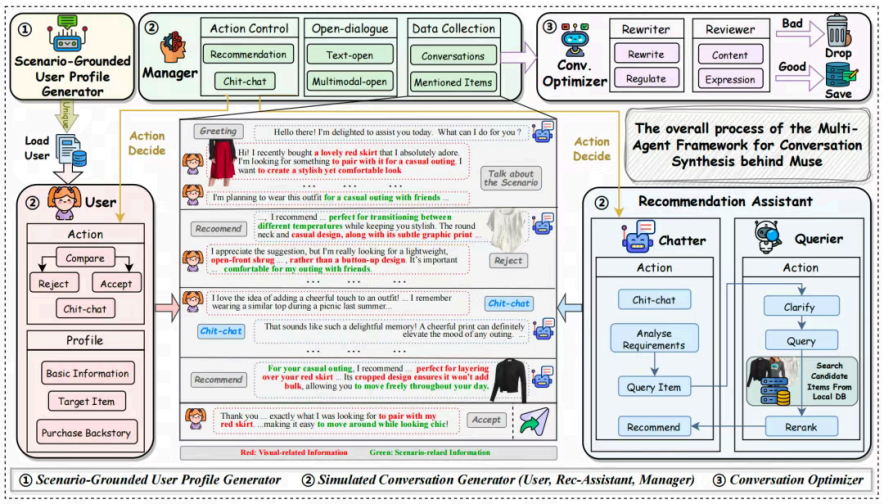
- 基本信息

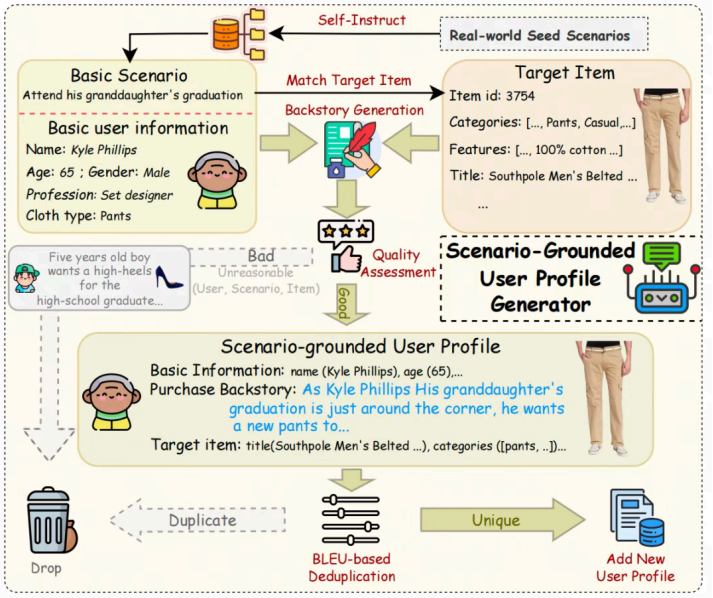
特点	数量	场景
大模型生成对话，基于真实场景和VLM生成的用户画像生成	7k个case，8.3w个对话	对话推荐数据集 服装领域



- 数据集构建

1. 用户画像生成器：收集多样的真实场景，生成用户画像，以便更好地匹配用户需求与产品特性。
2. 模拟对话生成器：利用用户画像进行角色扮演，模拟用户与推荐助手之间的对话。
3. 对话优化器：通过重写和审查机制提升对话的多样性和质量。





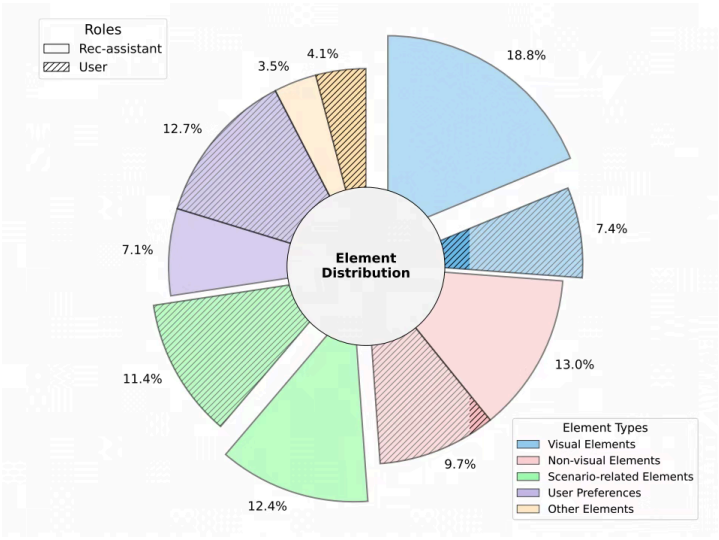


Figure 7: Distribution of mentioned products in MUSE.

- 数据集示例

"Basic Information": "Name: Timothy Barnett; Gender: male; Age: 40; Profession: Chief Strategy Officer"

"Purchase Backstory": "Timothy Barnett was buzzing with excitement as the date of the book launch approached. As the Chief Strategy Officer of a thriving start-up, he often mingled with influential authors and entrepreneurs at such events. This time, however, it was personal. The book was authored by his mentor, and he wanted to make a lasting impression while also supporting someone who had shaped his career. In the days leading up to the launch, Timothy found himself pondering how to present himself. He wanted to look professional yet trendy, an embodiment of his modern approach to business. He imagined himself surrounded by peers, exchanging business cards fluidly. However, the thought of his cards getting damaged or possibly even stolen filled him with dread. He needed something that would not only protect his cards but also reflect his stylish persona. \n\nAs he browsed through various online platforms, he stumbled upon a sleek, stainless steel card holder that promised to shield his cards from unwanted scanning. The idea of having RFID protection made him feel more secure, especially knowing that he would be in crowded spaces at the venue. The thought of confidently handing out his cards, knowing they were safeguarded, elevated his anticipation for the event. He could already picture himself networking effortlessly, showcasing both his credentials and his flair for contemporary style. Timothy felt this was exactly what he needed to enhance his experience at the book launch, and he decided it was time to make a purchase."

"Target Item":Title: Leopardd RFID Blocking Stainless Steel Card Holder Case (001) . ...

Figure 13: Two cases of the Scenario-Grounded User Profiles

Assistant: Hello! How may I assist you today?

User: I recently picked up a super comfy tunic, and I'm thrilled to style it. I'm in search of ideas for pieces that would enhance its chill vibe.

Assistant: What an exciting idea! A lively pattern would definitely inject some personality into your ensemble, complementing the solid-toned tunic beautifully and creating a visually appealing contrast."

User: Definitely! I'm currently exploring leggings with a festive, eye-catching holiday pattern. These vivid designs would create a stunning contrast when paired with the tunic.

Assistant: That's a delightful pairing! These festive printed leggings will undoubtedly inject some vibrant spirit into your comfortable tunic. Could you share the specific event or setting where you're planning to showcase this stylish ensemble?

User: I'm thinking of surprising Sarah with this festive ensemble for the holidays. My goal is to pick something that strikes a perfect balance between chic and cozy, allowing her to comfortably mingle with family and playfully dance with her little ones. It's crucial that the leggings feature a flexible waistband to ensure she can move around freely without any restrictions.

Assistant: If you are in search of the perfect pair to complement your tunic, the VIV Collection Plus Size Print Fashion Leggings are an excellent choice! Featuring a whimsical design of festive Christmas trees and ornaments, these leggings add a fun element to your look. Additionally, their flexible waistband ensures comfort, enabling your friend to enjoy holiday festivities and family gatherings with style and ease.

User: I see what you're getting at, but the festive design doesn't quite hit the mark for me. What I'm after are the Leggings with fun, adaptable vibe that would work perfectly for my friend's wardrobe, no matter the season.

Assistant: For a highly adaptable choice that will enhance your friend's cozy tunic, I suggest the Leggings Mania Regular/Plus Solid Buttery Soft High Waist Leggings. With their incredibly smooth texture and flexible waistband, these leggings provide exceptional comfort and mobility, ideal for family events and casual daily activities. Their versatile solid color enables effortless mixing and matching across various seasons!

User: Thanks for the suggestion, but I need to pass on those leggings since they don't provide any material details, which makes it hard to evaluate their quality. I need Leggings made from 100% polyester, really stand out for their superior softness and comfort, which is essential for what I'm looking for.

Assistant: Hey there! I've got a killer recommendation for your friend's wardrobe - the Two Left Feet Women's Holiday Leggings. These bad boys are super versatile and totally fun! Crafted from pure polyester, they rock a high-waist cut with this adorable quirky design that'll totally jazz up her tunic. The elastic waistband means maximum comfort, and trust me, they're soft as can be. Whether it's a family get-together or just a casual day out, these leggings will be her go-to piece all year long!

User: Thank you for the recommendation! I believe the Two Left Feet Women's Holiday Leggings are perfect for my friend. The quirky design adds a fun touch, and the 100% polyester material ensures they will be soft and comfortable for family gatherings and everyday wear. Plus, the high-waist and elastic waistband will provide the flexibility she needs to move around easily.

Figure 14: Text-open conversation case.

Assistant: Greetings! What assistance can I provide for you today?

User: I'm truly enthusiastic about my mentor's imminent book release and eager to connect with industry professionals. However, I'm concerned about potential risks to my professional networking materials, specifically the potential damage or theft of my business cards during the event.

Assistant: What an exciting idea! A lively pattern would definitely inject some personality into your ensemble, complementing the solid-toned tunic beautifully and creating a visually appealing contrast.

User: I'm seeking a business card design that effectively blends professional sophistication with contemporary style. My goal is to make a strong impression and feel self-assured when networking in a bustling environment.

Assistant: Let me suggest a sophisticated leather business card wallet that will elevate your professional image. This premium cowhide accessory offers ample storage for over 50 business cards, complemented by a practical ID compartment and extra slots for credit cards and cash. Its sleek design ensures your networking materials remain immaculate and easily accessible during your professional engagements."

User: I value your recommendation, but my preference leans towards an RFID-blocking card wallet. The aesthetic matters significantly to me - I'm seeking a slim, metallic accessory that not only shields my cards from potential electronic theft but also complements my style for the upcoming event.

Assistant: For the networking event ahead, I suggest the HAPTIME RFID Blocking Credit Card Holder. This elegant stainless steel accessory offers a chic appearance while safeguarding your cards against potential electronic theft. Its compact form factor allows you to carry multiple cards effortlessly, making it an ideal companion for professional networking situations. The integrated RFID blocking technology provides added security, enabling you to focus on making meaningful connections without worrying about card safety.

User: I'm captivated by the prospect of a streamlined, metallic card holder design! The memory of a tech conference floods back, where professionals flaunted their distinctive accessories that were both stylish and functional. It's fascinating how a compact item like a card holder can significantly enhance one's professional image during networking events. The sleek metallic aesthetic speaks volumes about personal branding, and the added RFID protection is a brilliant feature that offers peace of mind in bustling, crowded environments where security is paramount. Such an accessory isn't just about storage, it's a statement of sophistication and practicality.

Assistant: Absolutely, tech conferences are indeed a hub of creativity and style! The vibrant environment allows your individual flair to stand out while ensuring your belongings are safe. The HAPTIME RFID Blocking Credit Card Holder would perfectly enhance your outfit, providing both elegance and protection ideal for making a strong impression at your upcoming book launch without sacrificing security.

User: While I appreciate the recommendation, I'm specifically seeking an item with a contemporary and trendy design. The aesthetic appeal is crucial for me, and the suggested product's standard black appearance doesn't align with the sophisticated style I'm aiming for at the event.

Assistant: I recommend the Leopardd RFID Blocking Stainless Steel Card Holder Case for those seeking a contemporary and chic accessory. This sophisticated case boasts a streamlined, elegant appearance that elevates your personal style while offering top-tier protection against digital theft. Its sturdy build and convenient mechanism make it an ideal companion for professional gatherings, effectively safeguarding your cards from potential electronic scanning in busy settings.

User: I'm thrilled that the Leopardd RFID Blocking Stainless Steel Card Holder resonates so perfectly with my aesthetic and practical needs. Its elegant, streamlined profile not only complements my personal style but also offers robust security for my cards. The blend of contemporary design and utility makes it an exceptional choice for professional events like my mentor's book launch, ensuring I look sophisticated while keeping my cards well-protected.

Figure 15: Multimodal-open conversation case.

- 对数据集质量的评估

将 MUSE 与四个数据集（MMCONV、Redial、Inspired 和 P EARL）比较。采用 LLM 评估，随机抽取 200 个对话评估，涵盖对话自然性、逻辑连贯性、信息丰富性、产品上下文相关性和图像文本一致性等五个维度。

Table 3: Comparison of different datasets across multiple evaluation metrics

Metrics	MMCONV	Redial	INSPIRED	PEARL	MUSE
Natural(0-2)	1.41	1.57	1.71	1.66	**1.85**
Logical(0-2)	1.41	1.60	1.62	1.78	**1.88**
Informative(0-2)	1.37	1.53	1.51	1.67	**1.80**
P-C Correlation(0-2)	1.68	1.92	1.83	1.50	**1.98**
I-T Correspondence(0-2)	1.35	-	-	-	**1.91**

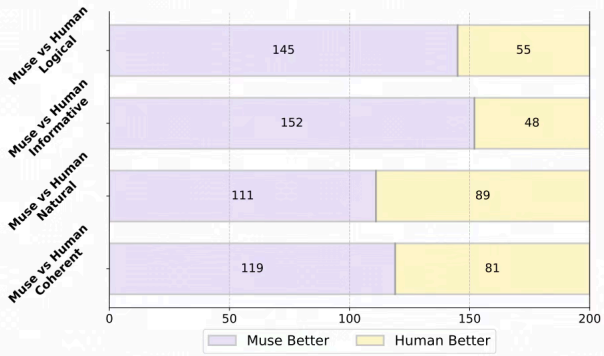


Figure 5: Utterance-level comparison: the quality between human responses and Muse's utterances,

基于MUSE数据集，比较不同模型的效果：

Table 4: Recommendation performance of different models under zero-shot and fine-tuning settings

Setting	Recall@10	Recall@20	MRR@10	MRR@20
LlaVA-NEXT-Llama3-8B				
Zero-Shot	0.16	0.25	0.07	0.09
Finetune	0.25	0.37	0.13	0.16
Yi-VL-6B				
Zero-Shot	0.15	0.23	0.07	0.08
Finetune	0.25	0.35	0.12	0.14
Qwen-2-VL-7B				
Zero-Shot	0.20	0.33	0.12	0.13
Finetune	**0.34**	**0.45**	**0.22**	**0.24**

Table 5: Response performance of different models under zero-shot and fine-tuning settings (p-value).

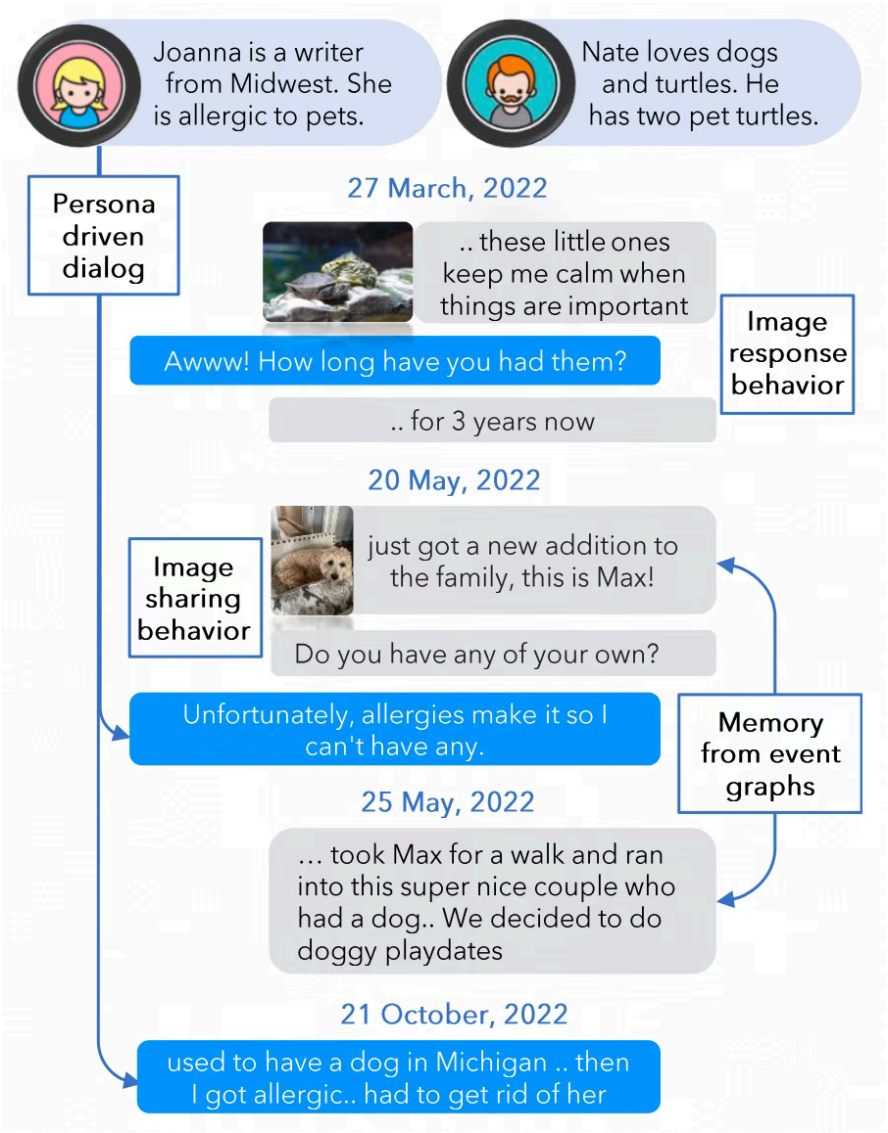
Setting	BLEU-4	ROUGE-1	ROUGE-L	Distinct-4	Avg. Words
LlaVA-NEXT-Llama3-8B					
Zero-Shot	17.1	17.2	9.61	0.63	86.6
Finetune	44.0	37.8	27.2	0.64	52.6
Yi-VL-6B					
Zero-Shot	16.5	16.7	9.48	0.58	107.7
Finetune	44.1	37.7	27.1	0.63	54.9
Qwen-2-VL-7B					
Zero-Shot	41.1	35.3	23.5	0.69	72.3
Finetune	**46.8**	**42.7**	**31.5**	**0.71**	**48.7**

LOCOMO

《Evaluating Very Long-Term Conversational Memory of LLM Agents》

- 基本信息

特点	数量	场景
大模型生成对话，基于个性化角色和时间事件图来构建对话	50个对话，每个对话平均300轮、9000个标记	评估LLM处理长对话的记忆能力：问题回答、事件总结和多模态对话生成



- 数据集构建

1. 人物设定与时间事件图：获取初始人物设定，利用 LLM扩展设定；每个agent构建的时间事件图包含多个事件，通过因果关系相互连接
2. 反思与回应机制：每次会话结束，生成总结，存储为短期记忆；每次对话的单个回合，作为观察内容，存储为长期记忆，在未来的对话中回忆。
3. 人工验证与编辑：通过人工矫正，消除长期不一致性、移除不相关图像，并确保对话内容与事件图对齐。人工对15%的对话回合进行了编辑，对19%的图像进行了替换或移除。

> 多模态对话生成任务中的“观察”，指从对话历史中检索到的与说话者相关的具体信息或事件描述，包含说话者的经历、个性特征、行为模式或与对话内容相关的上下文信息。
> 人类标注者对生成的对话进行编辑和验证，以确保长期一致性。具体包括以下任务：(1) 编辑对话以消除长期不一致之处；(2) 移除或替换无关的图像；(3) 验证和编辑对话内容与事件图之间的对齐性。标注者大约编辑了15%的对话回合，并且移除了或替换了大约19%的图像。

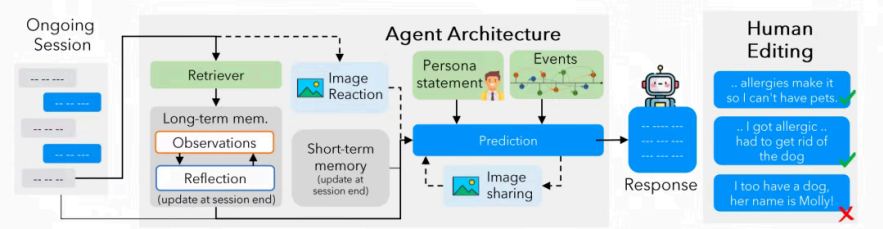


Figure 3: **Overview of the generative pipeline for LoCoMo.** Each LLM agent is assigned a distinct persona and a timeline of causally connected events in their file. The agent is equipped with a memory and reflection module to retrieve relevant history for dialog generation and is also enabled for image-sharing and image-reaction behaviors (left). The generated conversations are edited by human annotators to maintain long-range consistency (right).

Figure 11: **Example of edits made by annotators.** Human annotators are instructed to make edits in the LLM-generated conversations to remove irrelevant The prompt used to generate complete personas for the LLMs in our conversation generation pipeline (top) and examples of personas present in the LoCoMo dataset.

- 数据集指令

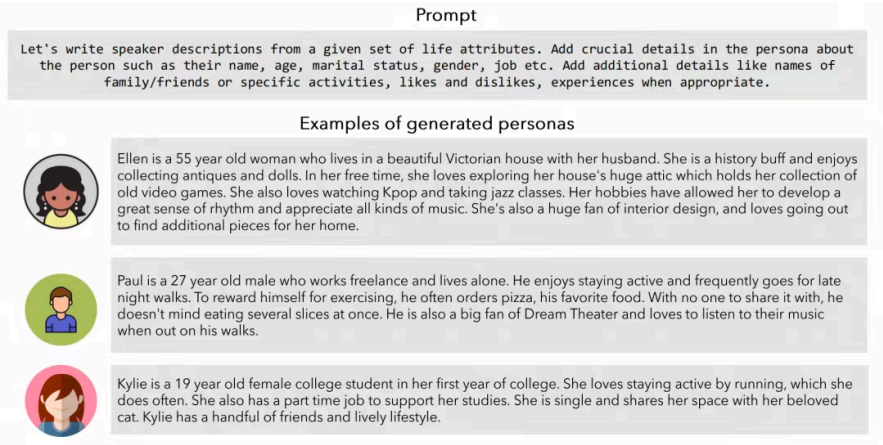


Figure 5: **Prompt for persona statement (p) generation and examples of personas in LoCoMo.** The prompt used to generate expanded persona statements (p) from initial personas (p_e) for the virtual agents in our conversation generation pipeline (top) and select examples of persona statements present in the LoCoMo dataset.

Base prompt for event graph generation

```
Let's write a graph representing events that occur in a person's life based on a short summary of their
personality. Nodes represent events and edges represent the influence of past sub-events on a current event.
- The graph is represented in the form of a json list.
- Each entry is a dictionary containing the following keys: "event", “date", "caused_by", "id".
- The "event" field contains a short description of the event.
- The “date" field contains a date.
- The "id" field contains a unique identifier for the event.
- The "caused_by" field represents edges and is a list of "id" of existing events that have caused this event.
Events in the "caused_by" field should occur on dates before the event they have caused. Generate as many
causal connections as possible.
- An example of a causal effect is when the event "started a vegetable garden" causes "harvested tomatoes".
- Events can be positive or negative life events.
```

Additional prompt for event graph initialization

```
For the following input personality, generate three independent events E1, E2 and E3 aligned with their
personality. Events can be positive or negative life events and should reflect evolution in the person's
relationships, state of mind, personality etc.
```

Additional prompt for iterative generation of causal events

```
For the following input personality, generate new events that are caused by one or more EXISTING events. Events
can be positive or negative life events and should reflect evolution in the person's relationships, state of
mind, personality etc. Do not repeat existing sub-events. Start and end your answer with a square bracket.
```

Figure 6: **Prompts for temporal event graph generation.** The prompt used to generate complete personas for the LLMs in our conversation generation pipeline (top) and examples of personas present in the LoCoMo dataset.

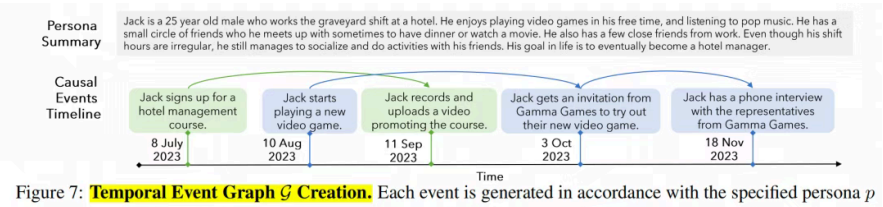


Figure 7: **Temporal Event Graph $\mathcal{G}$ Creation.** Each event is generated in accordance with the specified persona p and causal connections l between events are depicted to illustrate the casual relationships among them.

Prompt for generating summaries

```
In previous interactions, {previous_summary}. The current time and date are {current_date_and_time}.
{speaker_1_name} and {speaker_2_name} talked today and had the following conversation: {session} Summarize the
interactions between {speaker_1_name} and {speaker_2_name} so far. Include key details about both speakers and
include time references wherever possible.
```

Session (input)

1:14 pm on 25 May, 2023

Hey Nate! Haven't talked in a few days. Crazy things happened to me!

Hi Joanna! Long time no see! What's been going on? You sound excited!

Woo! I finally finished my first full screenplay and printed it last Friday. I've been working on for a while, such a relief to have it all done!

Wow, that sounds awesome! What's it about? Glad it's all down! Thanks, Nate! It's a mix of drama and romance!

Thanks, Nate! It's a mix of drama and romance!

Wow, that's amazing! How do you feel now that it's finished? Do you have any new plans for it?

I'm feeling a rollercoaster of emotions - relief, excitement, some anxiety - over finishing this project.

...

Previous summary (input)

On 21 January 2022, Joanna and Nate reunited after a long time without seeing each other. Nate won his first video game tournament playing Counter-Strike: Global Offensive. Joanna enjoys writing, reading, watching movies, and exploring nature as hobbies. They both share a love for movies, with Nate favoring action and sci-fi while Joanna prefers dramas and romcoms. Joanna recommended a romantic drama to Nate, which he expressed interest in watching. Nate praised Joanna's recommendation and promised to give it a try.

New summary (output)

Joanna and Nate reunited on 21 January 2022 after a long time without seeing each other. They share a love for movies, with Nate favoring action and sci-fi while Joanna prefers dramas and romcoms. Joanna recommended a romantic drama to Nate, which he expressed interest in watching. On 23 January 2022, at 2:01 pm, Joanna shared her excitement about finishing her first full screenplay, which is a mix of drama and romance, and her plans to submit it to film festivals. Nate congratulated her and shared his own experience with his pet turtles, recommending having pets for times of stress. Joanna mentioned her allergies to most animals with fur, prompting Nate to inquire further about her allergies and express empathy towards her situation. Despite her allergies, Joanna finds joy in writing and spending time with friends. The conversation ended on a positive note with Nate encouraging Joanna to keep pursuing her passion for writing.

Figure 8: **Prompt for generating conversation summaries.** The prompt used to iteratively generate a summary for the current session by conditioning on summary from preceding sessions and the raw conversation logs of the current session (top); and an example of inputs for the prompt and corresponding output summary of a session from the LoCoMo dataset.

Prompt for generating observations

```
Write a concise and short list of all possible OBSERVATIONS about each speaker that can be gathered from the
CONVERSATION. Each observation should contain a piece of information about the speaker. The OBSERVATIONS should
be objective factual information about the speaker that can be used as a database about them. Avoid abstract
observations about the dynamics between the two speakers such as 'speaker is supportive', 'speaker appreciates'
etc. Do not leave out any information from the CONVERSATION.
```

Conversation (input)

1:56 pm, May 8, 2023

Hey Joanna! Long time no see! What's up? ..

Hey Nate! Long time no see! I've been working on a project lately - it's been pretty cool. What about you - any fun projects or hobbies?

I won my first video game tournament last week - so exciting!

Wow Nate! Congrats on winning! Tell me more - what game was it?

Thanks! it's a team shooter game.

Wow, great job! What was is called?

The game was called Counter-Strike: Global Offensive, and me and my team had a blast to the very end!

Sounds like a fun experience .. if I'm not into games.

...

...

Observations (output)

Joanna

- Joanna has been working on a project recently.
- Joanna enjoys writing, reading, watching movies, and exploring nature as hobbies.
- Joanna is into dramas and romcoms when it comes to movies.
- Joanna recommends a romantic drama movie that is all about memory and relationships.
- Joanna watched the recommended movie around 3 years ago and even owns a physical copy.

Nate

- Nate won his first video game tournament last week.
- The video game Nate won the tournament in is called Counter-Strike: Global Offensive.
- Playing video games and watching movies are Nate's main hobbies.
- Nate enjoys action and sci-fi movies.
- Nate loves watching classics.

Figure 9: **Prompts for generating observations from conversations.** The prompt used to generate observations from a conversation (top); and an example of inputs for the prompt and corresponding output observations for a session from the LoCoMo dataset.

Image sharing: Prompt for generated image caption → image query

```
Let's write short image search queries from textual descriptions of photos shared by a user.
Queries should not include names of people, years and other irrelevant details. For example:

Input: That sounds relaxing, Jeremy! As for video game suggestions, have you ever tried "The Legend
of Zelda: Breath of the Wild"? It's an open-world adventure game that I absolutely love. [shares a
photo of Link standing in front of a breathtaking landscape] Have a look at this stunning view!
Output: the legend of zelda: breath of wild link landscape

Input: {generated_image_caption}
Output:
```

Image response: Prompt for shared image caption → response grounded in image

```
{speaker_1_persona}
{speaker_2_persona}

{speaker_1_name} says, {current_turn}, and shares a photo of {shared_image_caption_blip2}. Write
the most natural question or comment {speaker_2_name} can include in their response.
```

Figure 10: **Prompts for image-sharing and image-response behavior.** The prompt used to convert a caption generated by the virtual agent into an image query for the web-based image crawler in our pipeline (top), and the prompt used to generate a response grounded in the image shared by a virtual agent during a conversation as well as the personas of the respective speakers (bottom).

- 对模型能力的评估

1. 问答任务能力：具有有限上下文长度的LLM，对极长对话的理解较差。gpt-4-turbo表现最佳（32.4），但仍显著低于人类基准（87.9）。长上下文LLM能理解更长的叙述，但容易产生幻觉。RAG将对话存储为观察内容时有效，随着检索观察数量的增加，性能改善减弱。
2. 事件总结任务能力：事件总结任务要求模型理解多次会话中事件的时间和因果关系。gpt-3.5-turbo召回率和F1分数最高。长上下文模型对于上下文的利用不足。五类主要错误，包括信息缺失、幻觉、对话线索误解、说话者归属错误以及不重要对话被错误视为重要事件。
3. 多模态对话生成任务结果：包含上下文的训练提升了生成性能，将观察内容作为上下文时，结果更显著。“观察内容”包含说话者的经历信息，导致生成的对话和图像更符合说话者的个性。
4. 总体效果：LLMs在理解长时间叙述和提取时间及因果关系方面存在困难。尽管长上下文LLM和RAG提供了缓解，但LLM仍显著落后于人类，尤其是在时间推理和开放领域知识问题上。

Figure 2: **Overview of our evaluation framework**. We propose three tasks: question answering, event summarization and multimodal dialog generation to evaluate models' comprehension in very long-term dialogues.

Category	Model	Context Length	Answer Prediction (F1)					
			Single Hop	Multi Hop	Temporal	Open Domain	Adversarial	Overall
Human	Human	-	95.1	85.8	92.6	75.4	89.4	87.9
Base	Mistral-Instruct-7B	8K	10.2	12.8	16.1	19.5	17.0	13.9
	Llama-2-Chat-70B	4,096	19.7	14.4	13.3	15.9	22.1	17.9
	GPT-3.5-turbo	4,096	**29.9**	23.3	**17.5**	**29.5**	12.8	22.4
	GPT-4-turbo	4,096	23.4	**23.4**	10.4	24.6	**70.2**	**32.1**
Long context	GPT-3.5-turbo-16K	4K	31.7	25.4	16.8	27.6	**13.1**	24.1
		8K	38.8	31.2	21.0	35.0	8.4	25.2
		12K	51.1	40.4	**25.0**	36.5	6.4	33.5
		16K	**56.4**	**42.0**	20.3	**37.2**	2.1	**37.8**

Table 2: **Question answering performance** of Base and Long-context models. Optimal performance is in **bold**. Results are based on F1-score for answer prediction; higher is better.

Retrieval Unit	top-k	Answer Prediction (F1 score)						Recall Accuracy (R@k)					
		Single Hop	Multi Hop	Temporal	Open Domain	Adver--sarial	Overall	Single Hop	Multi Hop	Temporal	Open Domain	Adver--sarial	Overall
None	-	29.9	23.3	17.5	29.5	12.8	22.4	-	-	-	-	-	-
Dialog	5	42.9	19.4	21.3	35.8	31.9	31.7	66.2	34.4	89.2	38.5	45.7	58.8
	10	46.3	26.8	24.8	37.5	**29.8**	34.6	72.8	247.4	97.3	53.8	54.3	67.5
	25	48.1	36.1	**26.2**	**43.4**	23.4	**35.8**	87.5	64.1	**97.3**	**67.9**	69.1	79.9
	50	**50.9**	**37.2**	24.6	38.3	17.0	34.8	**90.4**	**75.5**	97.3	67.9	**77.7**	**84.8**
Observation	5	44.3	30.6	41.9	40.2	**44.7**	**41.4**	52.9	40.1	81.1	38.5	29.8	49.6
	10	42.2	30.5	**42.1**	41.9	36.2	38.8	57.4	53.1	**83.8**	46.2	41.5	57.1
	25	**44.6**	33.2	41.8	41.9	27.7	38.0	**71.3**	63.8	83.8	66.7	45.7	66.0
	50	44.0	**34.5**	41.1	**41.9**	27.7	37.8	72.8	**73.2**	83.8	**74.4**	**56.4**	**71.1**
Summary	2	34.6	15.7	26.9	26.5	36.2	29.9	68.4	39.6	56.8	50.0	73.4	61.5
	5	**36.6**	**16.6**	**31.0**	**34.7**	38.3	**32.5**	81.6	57.0	70.3	60.3	86.2	75.1
	10	34.5	14.7	29.3	31.6	**40.4**	31.5	**93.4**	**82.3**	**91.9**	**80.8**	**94.7**	**90.7**

Table 3: **Question answering performance** of RAG-based GPT-3.5-turbo-16k. Optimal performance is in **bold**. Results are based on F1-score metric for answer prediction and recall@k for recall accuracy; higher is better.

Category	Model	Context Length	ROGUE			FactScore		
			ROGUE-1	ROGUE-2	ROGUE-L	Precision	Recall	F1
Base	Mistral-Instruct-7B	8K	29.4	7.2	14.1	27.1	19.8	23.0
	Llama-2-Chat-70B	4,096	28.1	9.3	14.8	36.3	22.7	28.3
	GPT-4-turbo	4,096	38.8	11.4	20.6	**51.6**	41.8	45.1
	GPT-3.5-turbo	4,096	**41.1**	**13.5**	**20.9**	45.3	**46.5**	**45.9**
Long context	GPT-3.5-turbo-16K	16K	36.2	8.5	16.4	42.3	37.8	39.9

Table 4: **Event summarization performance** of Base and Long-context models. The optimal performance is shown in **bold**. Results are based on ROUGE and FactScore (Min et al., 2023) metrics; higher is better.

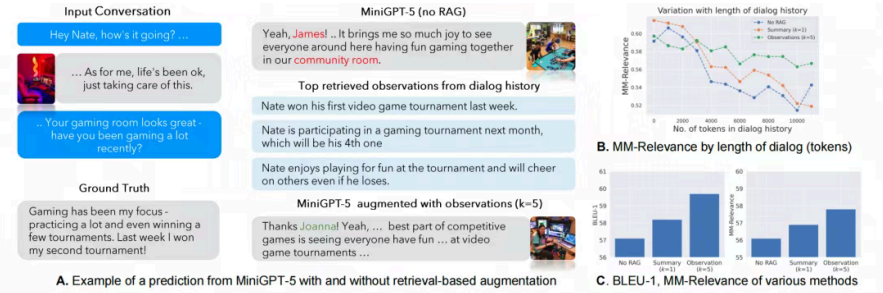


Figure 4: **Multimodal dialog generation performance of MiniGPT-5**. (A) an example of multimodal dialog predicted using MiniGPT5 with and without observation as retrieved context, (B) Variation of MM-Relevance score with length of dialog history, and (C) comparison of RAG-based MiniGPT-5 methods.

评测实验配置：

1. 问答任务（Question Answering）

- 模型类型：评估了三种类型的模型：
  - （1）基础LLM（Base LLMs）：使用受限的上下文长度，早期对话被忽略，包括Mistral-7B、LLaMa-70B-chat、gpt-3.5-turbo和gpt-4-turbo。
  - （2）长上下文LLM（Long-context LLMs）：扩展的上下文窗口，包括gpt-3.5-turbo-16k。
  - （3）检索增强生成（RAG）：从对话历史、观察（关于说话者的断言）或会话级总结中检索相关上下文，使用DRAGON作为检索器，gpt-3.5-turbo-16k作为阅读器。
- 实验细节：对于图像内容，实验中用图像的标题替代原始图像（Li et al., 2023b），并使用最先进的LLM对交织在对话中的纯文本和图像标题进行推理。多模态对话生成任务中直接使用图像。

2. 事件总结任务（Event Summarization）

- 模型类型：使用问答任务中的基础和长上下文设置，但不包括RAG，因为总结任务需要对整个对话有全面理解，而非仅检索特定部分内容。
- 实验细节：实现增量总结，即迭代创建先前会话的总结，并以此作为后续会话总结的基础（Chang et al., 2023）。

3. 多模态对话生成任务（Multi-modal Dialogue Generation）

- 模型训练：使用自动生成的50个对话（没有人工过滤）作为训练数据，训练三种版本的MiniGPT-5：
  - （1）基础模型（Base）：仅训练于先前的对话轮次。
  - （2）+总结（+ summary）：训练于先前的对话轮次和全局对话总结。
  - （3）+观察（+ observation）：训练于先前的对话轮次和从对话历史中检索到的观察。
- 实验细节：每个模型运行均以MMDialog（Feng et al., 2023）微调的MiniGPT-5检查点初始化。

04 Memory Evaluation

MemoryAgentBench

《Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions》

- 基本信息

数据集part1	数据集part2	场景
重构多个现有数据集	并引入新数据集EventQA和FactConsolidation	评估LLM的记忆能力：准确检索、测试时学习、长程理解和冲突解决

LLM-Agent的评估主要集中在推理、规划和执行能力上，而记忆能力的评估则相对不足。记忆能力包括如何记忆、更新和检索长期信息。

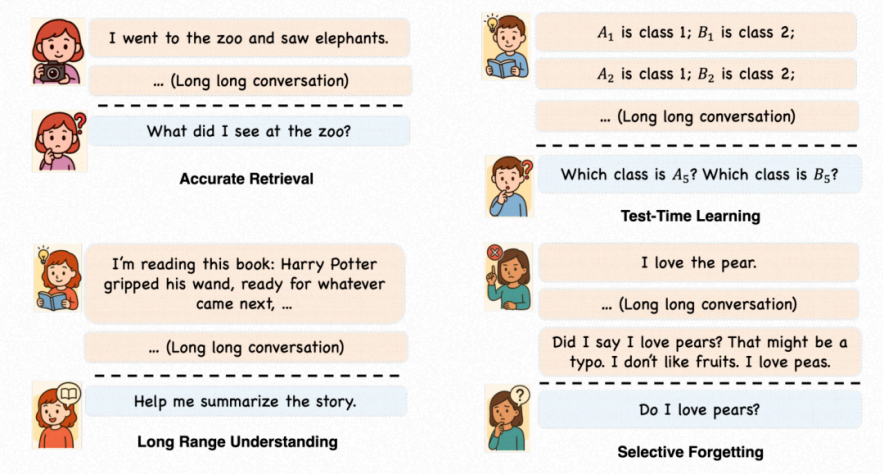
本文识别出四个对记忆代理至关重要的核心能力：

1. 准确检索（Accurate Retrieval, AR）：指agent从长对话历史中识别并检索重要信息的能力。

2. 测试时学习（Test-Time Learning, TTL）：指agent动态获取新技能的能力，而无需额外训练。类似于LLM)的上下文学习，通过对话历史中的少量示例学习新任务。

3. 长程理解（Long-Range Understanding, LRU）：指agent在长对话中形成抽象的、高层次理解的能力，能够回答需要整体理解的问题。

4. 冲突解决（Conflict Resolution, CR）：指agent在面对新旧信息冲突时，检测并解决矛盾的能力。agent能够识别并舍弃过时或错误的信息，以反映最新的现实状态。



- 数据集构建

1. 本文识别出四个对记忆代理至关重要的核心能力：准确检索（Accurate Retrieval）、测试时学习（Test-Time Learning）、长程理解（Long-Range Understanding）和冲突解决（Conflict Resolution）。
2. MemoryAgentBench结合了重新构建的现有数据集，并引入新数据集EventQA和FactConsolidation，分别用于评估准确检索和冲突解决能力。
3. 提供了一个统一的评估框架，允许不同类型的记忆代理在相同的条件下进行比较，从而揭示其在记忆能力上的差异。
4. 本文评估了三种主要类型的记忆代理：长上下文代理、检索增强生成（RAG）代理和代理记忆代理，揭示了当前方法在处理动态记忆更新和长程一致性任务时的局限。

Limitation：本文指出，当前使用的合成数据集可能无法完全反映真实用户对话的特征，未来的工作将致力于收集和整理更具现实性的真实数据集。

> 评估了三种主要类型的记忆代理：
> 长上下文代理（Long-Context Agents）： 现代语言模型通常支持扩展的上下文窗口（从128K到超过1M tokens）。一种直接的策略是维护一个最近token的上下文缓冲区。
> 检索增强生成（RAG）代理（RAG Agents）： RAG代理通过将过去的信存储在外部记忆池中，并在需要时检索相关内容来解决上下文限制。
> 代理记忆代理（Agentic Memory Agents）： 代理记忆代理通过采用代理循环（agentic loops）来超越静态记忆存储，代理可能会重新表述问题、进行记忆查找并更新其工作记忆。定期更新的数据集池。

Table 1: Overview of evaluation datasets. We select datasets that cover various important long-context capabilities. In the table, we underline the datasets we constructed ourselves. AvgL.: Average Context Length (measured using the GPT-4o-mini model's tokenizer).

Category	Dataset	Metrics	AvgL.	Description
Accurate Retrieval	SH-Doc QA	Accuracy	197K	Single-Hop Gold passage retrieval QA.
	MH-Doc QA		421K	Multiple-Hop Gold passage retrieval QA.
	LongMemEval (S*)		355K	Dialogues based QA.
	EventQA		534K	Novel multiple-choice QA on characters events.
Test-time Learning	BANKING77	Accuracy	103K	Banking intent classification, 77 labels.
	CLINC150			Intent classification, 151 labels.
	NLU			Task intent classification, 68 labels.
	TREC Coarse			Question type classification, 6 labels.
	TREC Fine			Question type classification, 50 labels.
	Movie Recommendation	Recall@5	1.44M	Recommend movies based on provided dialogues examples.
Long Range Understanding	∞Bench-Sum	F1-Score	172K	Novel summarization with entity replacement.
	Detective QA	Accuaracy	124K	Long-range reasoning QA on detective novels.
Selective Forgetting	FactConsolidation-SH	Accuracy	262K	Single hop reasoning in facts judgment.
	FactConsolidation-MH			Multiple hop reasoning in facts judgment.

- 数据集指令

Prompts Used for Memory Construction on Various Tasks

***Document Question Answering* (*SH-Doc QA* or *MH-Doc QA*):**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context consists the documents I have read: ⟨chunk⟩. Please memorize it and I will ask some questions based on it in future.
⟨Assistant⟩: Sure! I have learned the documents and I will answer the question you ask.

***LME(S*)*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the dialogue history that I have talked with a ChatBoT: ⟨chunk⟩. Please memorize it and I will ask some questions based on it in future.
⟨Assistant⟩: Sure! I have memorized the dialogue history and I will answer the question you ask.

***EventQA*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the sections in a novel that I have read: ⟨chunk⟩. Please memorize it and I will ask some questions based on it in future.
⟨Assistant⟩: Sure! I have memorized the novel sections and I will answer the question you ask.

***Multi-Class Classification (MCC)*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the examples I have learned: ⟨chunk⟩. Please memorize it and I will ask some questions based on it in future.
⟨Assistant⟩: Sure! I have memorized the examples and I will answer the question you ask.

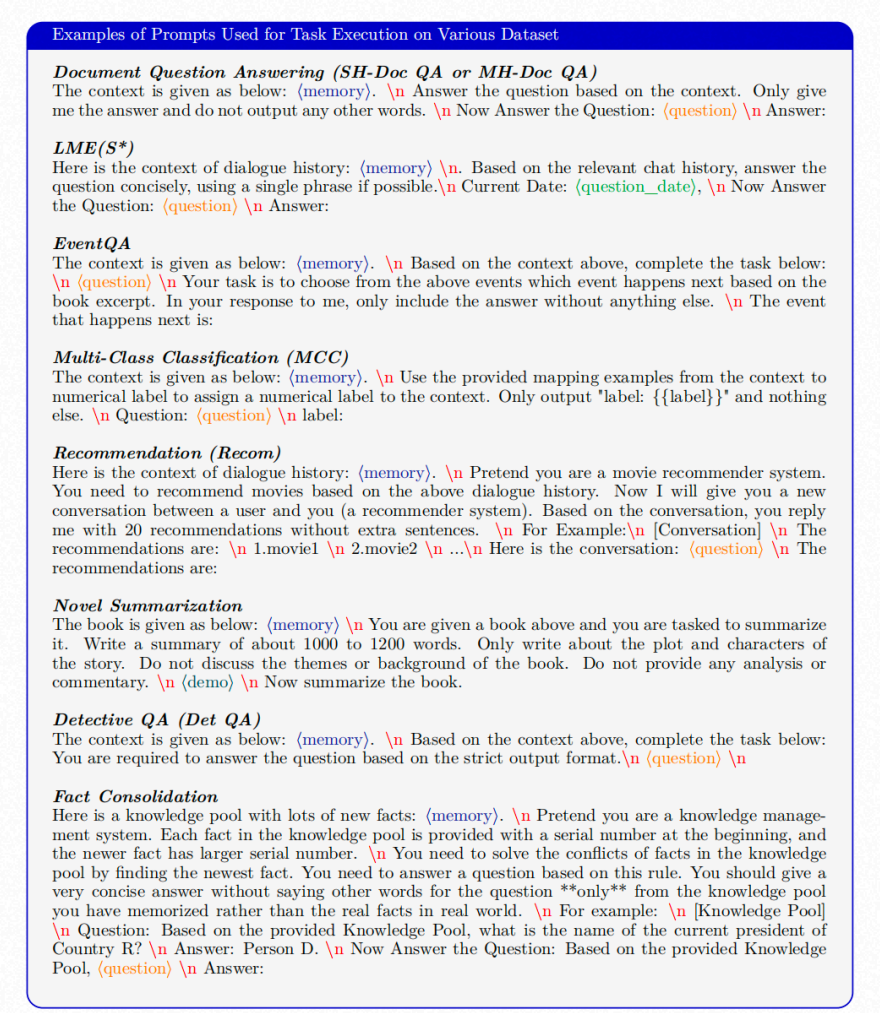
***Recommendation (Recom)*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the dialogue history that I have talked with a recommender system: ⟨chunk⟩. Please memorize it and I will ask some questions based on it in future.
⟨Assistant⟩: Sure! I have memorized the dialogues and I will answer the question you ask.

***Novel Summarization*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the sections in a novel that I have read: ⟨chunk⟩. Please memorize it and I will need you to summarize based on it in future.
⟨Assistant⟩: Sure! I have memorized the novel sections and I will summarize them when you ask.

***Detective QA (Det QA)*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the sections in a novel that I have read: ⟨chunk⟩. Please memorize it and I will need you to answer the questions based on the entire novel.
⟨Assistant⟩: Sure! I have memorized the novel sections and I will answer the questions you ask.

***Selective Forgetting (SF)*:**
Here is a dialogue between User and Assistant on ⟨time⟩:
⟨User⟩: The following context is the facts that I have learned: ⟨chunk⟩. Please memorize it and I will need you to answer the questions based on the order of facts.
⟨Assistant⟩: Sure! I have memorized the facts and I will answer the questions you ask.

Figure 4: The prompts we use for the agents to create the memory.



Examples of Prompts Used for Task Execution on Various Dataset

Document Question Answering (SH-Doc QA or MH-Doc QA)
The context is given as below: ⟨memory⟩. \n Answer the question based on the context. Only give me the answer and do not output any other words. \n Now Answer the Question: ⟨question⟩ \n Answer:

LME(S*)
Here is the context of dialogue history: ⟨memory⟩ \n. Based on the relevant chat history, answer the question concisely, using a single phrase if possible.\n Current Date: ⟨question_date⟩, \n Now Answer the Question: ⟨question⟩ \n Answer:

EventQA
The context is given as below: ⟨memory⟩. \n Based on the context above, complete the task below: \n ⟨question⟩ \n Your task is to choose from the above events which event happens next based on the book excerpt. In your response to me, only include the answer without anything else. \n The event that happens next is:

Multi-Class Classification (MCC)
The context is given as below: ⟨memory⟩. \n Use the provided mapping examples from the context to numerical label to assign a numerical label to the context. Only output "label: {{label}}" and nothing else. \n Question: ⟨question⟩ \n label:

Recommendation (Recom)
Here is the context of dialogue history: ⟨memory⟩. \n Pretend you are a movie recommender system. You need to recommend movies based on the above dialogue history. Now I will give you a new conversation between a user and you (a recommender system). Based on the conversation, you reply me with 20 recommendations without extra sentences. \n For Example:\n [Conversation] \n The recommendations are: \n 1.movie1 \n 2.movie2 \n ...\n Here is the conversation: ⟨question⟩ \n The recommendations are:

Novel Summarization
The book is given as below: ⟨memory⟩ \n You are given a book above and you are tasked to summarize it. Write a summary of about 1000 to 1200 words. Only write about the plot and characters of the story. Do not discuss the themes or background of the book. Do not provide any analysis or commentary. \n ⟨demo⟩ \n Now summarize the book.

Detective QA (Det QA)
The context is given as below: ⟨memory⟩. \n Based on the context above, complete the task below: You are required to answer the question based on the strict output format.\n ⟨question⟩ \n

Fact Consolidation
Here is a knowledge pool with lots of new facts: ⟨memory⟩. \n Pretend you are a knowledge management system. Each fact in the knowledge pool is provided with a serial number at the beginning, and the newer fact has larger serial number. \n You need to solve the conflicts of facts in the knowledge pool by finding the newest fact. You need to answer a question based on this rule. You should give a very concise answer without saying other words for the question **only** from the knowledge pool you have memorized rather than the real facts in real world. \n For example: \n [Knowledge Pool] \n Question: Based on the provided Knowledge Pool, what is the name of the current president of Country R? \n Answer: Person D. \n Now Answer the Question: Based on the provided Knowledge Pool, ⟨question⟩ \n Answer:

Figure 5: The example prompts we use for the *Memory Agents* in Table 2. Here ⟨memory⟩ refers to the accumulated text from the sequential inputs.

- 评估

1. RAG方法在准确检索任务中表现优越，RAG代理在准确检索类别的任务中优于GPT-4o-mini，RAG代理在提取回答问题所需的小段文本方面表现出色。
2. 其次，长上下文模型在测试时学习（TTL）和长范围理解（LRU）任务中表现最佳。RAG方法仅能从过去的上下文中检索部分信息，缺乏对输入的整体理解能力。
3. 所有现有方法在冲突解决（CR）任务上均表现不佳，尤其是在多跳场景中，准确率最高仅为6%。只有长上下文代理在单跳场景中取得了相对合理的结果。

LONGMEMEVAL

《LONGMEMEVAL: BENCHMARKING CHAT ASSISTANTS ON LONG-TERM INTERACTIVE MEMORY》

- 基本信息

特点	数量	场景
提出一种统一框架，优化记忆设计：会话分解、事实增强的键扩展和时间感知的查询扩展	LONGMEMEVAL-S （约115k个tokens）和LONGMEMEVALM （500个会话，约150万tokens）	评估聊天助手的长期记忆能力：信息提取、多会话推理、时间推理、知识更新和拒绝回答

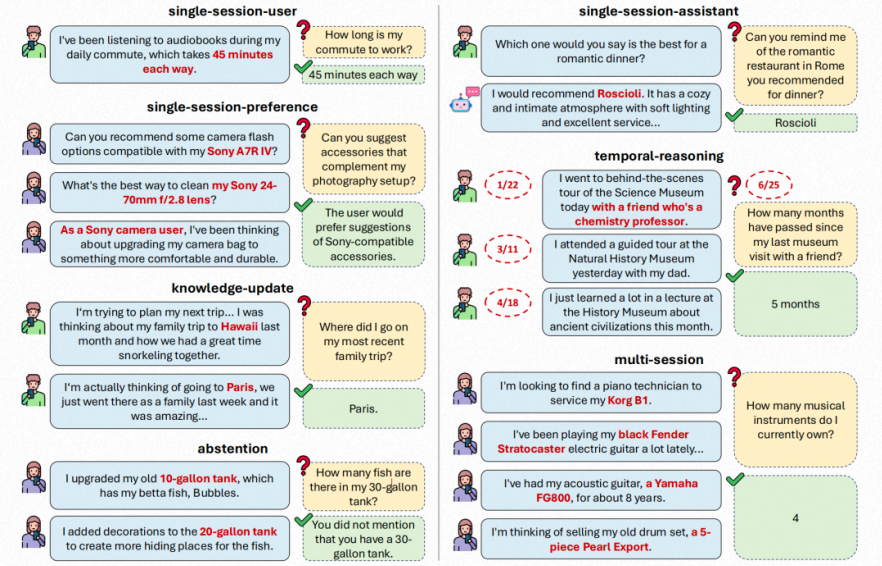
会话分解（Session Decomposition）：在长期记忆系统中，用户与助手的会话往往内容繁多且涵盖多个话题。将每个会话作为一个整体存储可能导致检索效率低下，而过度压缩会丢失细节信息，影响系统对复杂问题的回答能力。将会话分解为多个“轮次”，并进一步提取摘要、关键短语或用户事实。

事实增强的键扩展（Fact-Augmented Key Expansion）：传统方法使用会话或轮次内容作为键，可能无法有效捕捉到查询与记忆之间的关联。键扩展策略通过提取value中的摘要、关键短语、用户事实和时间戳事件来增强键。

时间感知的查询扩展（Time-Aware Query Expansion）：在涉及时间的查询任务中，单纯依赖相似性搜索可能无法满足需求。在索引阶段，从文本中提取时间戳事件；在检索阶段，从查询中推断时间范围，并据此过滤掉不相关的信息。

- 数据集构建

LONGMEMEVAL基准由500个问题组成。每个问题从一个或多个多轮任务导向的用户-助手对话中提取信息。两种配置：LONGMEMEVAL-S（约115k个tokens）和LONGMEMEVALM（500个会话，约150万tokens）。



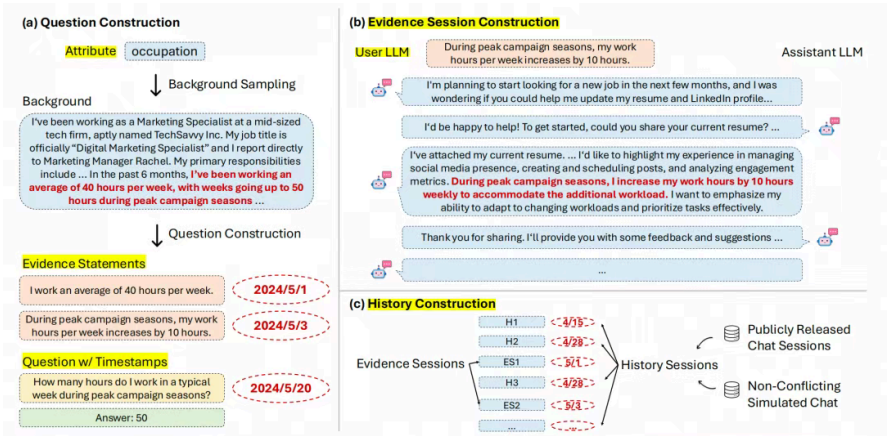


Figure 2: Data creation pipeline of LONGMEMEVAL. (a) Human experts construct all the questions and evidence statements. (b) Then, the evidence sessions are LLM-simulated and human-edited. (c) The full user-AI chat history is constructed at test time, whose length is freely configurable.

- 数据集指令

Table 5: Human-constructed user attribute ontology. Each attribute represents a unique dimension of human experience along which a user biography could be constructed. For LONGMEMEVAL, we sample user backgrounds based on each of the attributes and construct questions on top of them.

Attribute 1: Demographic Information
age, gender, ethnicity, nationality, language, education level, occupation
Attribute 2: Lifestyle
2.1: Shopping: online shopping frequency, favorite stores, loyalty program, sales events, coupons, gift purchasing habits, eco-friendly product preferences, luxury vs budget shopping, technology gadget purchasing, fashion and apparel, grocery shopping, shopping for others
2.2: Media Consumption: book, movie, tv show, music, podcast, video game, streaming service, theater, magazine and newspaper, youtube, educational content, audiobook and e-book
2.3: Social Media Engagement: posting, commenting, followers, groups, hashtags, campaigns, messaging, live streaming, social media breaks
2.4: Daily Routines: wake-up time, bedtime, work or school start time, meal time, exercise routines, coffee or tea break, commuting, evening activities, weekend routines, cleaning schedules, time spent with family or friends
2.5: Travel: frequency, destination, road trips, travel agencies, outdoor adventures, airlines, hotel, travel with family vs solo travel, packing habits
2.6: Recreation: reading, painting, musical instruments, dancing, watching sports, participating in sports, gardening, bird watching, fishing or hunting, board games, video games, fitness classes, yoga, sculpting, photography, stand-up comedy, writing, collecting, model building, aquarium keeping
2.7: Eating and Cooking: home cooking, food delivery, vegetarian or vegan, favorite cuisines, snacking habits, barbecue, baking, cocktail mixing, cooking classes
2.8: Event Participation: concerts, theater, galleries and museums, sports games, film festivals, religious services, book readings, charity events, trade shows, lectures or workshops, theme parks, local markets, networking events, sports, auto racing, workshops, museum tours
Attribute 3: Situational Context
3.1: Home: living room, kitchen, bathroom, room style, room lighting, furniture, technology, plants
3.2: Social Context: alone, family, friends, interactions with strangers
3.3: Time Context: time of day, day of week, seasonal
Attribute 4: Life Events
graduations, academic achievements, study abroad, significant academic projects, job promotions, starting a business, births and adoptions, marriages, family reunions, illness or surgeries, mental health journeys, purchasing a home, trips, movement, living abroad, refugee or immigration, loss of loved ones, name change, belief, milestone
Attribute 5: Belongings
cars, bikes, vehicles, computer, phone, pet, farm animal, animal care items, home, land, art, antiques, collectible, rare items, clothing, jewelry, shoes, bag, sports gear, musical instruments, health related devices, crafting, photography

I will give you a topic. Please imagine you are a user that wants to recall and record recent personal facts along the topic. Generate a long text describing these personal facts. Use your imagination and generate the personal facts. Make it long and involve several recent facts or recent events spanning many days, weeks, or monthes. You may state the facts in plain language and no need to make it story-like.

Topic: {attribute}

Recent Personal Facts related to {attribute}:

Figure 7: The prompt for constructing user backgrounds based on an attribute.

I will give you a past memory. Use the memory to act as a normal user to chat with a chat assistant. In the chat, you may ask it to assist you various tasks or ask it about various information. However, make sure that your convey the following fact about you: "{evidence_statement}". In addition, make sure your message is concise (1-2 simple sentences), since the real users often do not bother write a long message. I will provide you with the chat history and the response from the assistant. Directly generate the next response from the user's perspective. You must simulate the tone of a neutral user and do not be overly enthusiastic, verbose, formal, or polite. For conciseness, DO NOT react to the assistant's message with e.g., "thanks" or "I will do that". Instead, directly state the follow-up questions or new questions.

Memory: {background}

Chat History:

assistant: Hi! How can I assist you today?

... (more rounds as the conversation continues) ...

Figure 8: The prompt for instruction an LLM to act as a user and initiates a task-oriented dialogue with another LLM. Both the background and the evidence statement is provided. This prompt is used for the question type **single-session**. For the other question types, the prompt components are slightly different but the prompt overall follows the same style.

temp-reasoning

I will give you a question, a correct answer, and a response from a model. Please answer yes if the response contains the correct answer. Otherwise, answer no. If the response is equivalent to the correct answer or contains all the intermediate steps to get the correct answer, you should also answer yes. If the response only contains a subset of the information required by the answer, answer no. In addition, do not penalize off-by-one errors for the number of days. If the question asks for the number of days/weeks/months, etc., and the model makes off-by-one errors (e.g., predicting 19 days when the answer is 18), the model's response is still correct.

---

knowledge-update

I will give you a question, a correct answer, and a response from a model. Please answer yes if the response contains the correct answer. Otherwise, answer no. If the response contains some previous information along with an updated answer, the response should be considered as correct as long as the updated answer is the required answer.

---

single-session-preference

I will give you a question, a rubric for desired personalized response, and a response from a model. Please answer yes if the response satisfies the desired response. Otherwise, answer no. The model does not need to reflect all the points in the rubric. The response is correct as long as it recalls and utilizes the user's personal information correctly.

Other question types

I will give you a question, a correct answer, and a response from a model. Please answer yes if the response contains the correct answer. Otherwise, answer no. If the response is equivalent to the correct answer or contains all the intermediate steps to get the correct answer, you should also answer yes. If the response only contains a subset of the information required by the answer, answer no.

Figure 10: Evaluation instructions for the GPT-4o judge. We provide the question, answer, and the model's hypothesis after the instruction and ask GPT-4o to directly generate "yes" or "no".

- 评估

1. 评估过程中，采用了四个关键阶段的统一框架：索引、检索、读取和评估。在索引阶段，比较不同的值表示策略--会话分解和用户事实提取。在检索阶段，采用时间感知的查询扩展策略，以提高时间推理问题的准确性。在读取阶段，采样Chain-of-Note和结构化提示格式，以提升模型的读取准确性。
2. 现有的商业聊天助手和长上下文LLM在LONGMEMEVAL基准上表现不佳，准确率下降30%至60%。当前的记忆机制在处理复杂的长期交互时仍存挑战，需要更复杂的记忆机制，提高个性化和可靠性。

MemBench

《MemBench: Towards More Comprehensive Evaluation on the Memory of LLM-based Agents》

- 基本信息

特点	场景
包含事实记忆和反思记忆的多场景数据集，并引入参与和观察两种互动场景	评估LLM-agent记忆能力，多指标评估（准确率、召回率、容量和效率）

已有研究对LLM-Agent的记忆能力进行了评估，但这些评估局限性。首先，许多研究主要集中在事实记忆的评估上，而忽视了反思记忆的评估。其次，现有评估大多局限于参与场景，未能充分考虑观察场景的影响。此外，现有研究往往只关注记忆机制的有效性，而未考虑其效率和容量，这在实际应用中同样重要。

事实记忆（Factual Memory）：涉及用户或与其相关实体的事实属性，例如亲属的年龄或职业、事件的时间细节等。这些信息通常在用户与代理的日常对话中。通过提出有关这些属性的问题，测试代理的多种记忆能力，包括信息提取、跨会话推理、知识更新、时间推理等。

反思记忆（Reflective Memory）：指基于用户表达的低级偏好（例如具体的事实属性，如用户对某道菜的喜爱）和总结出的高级偏好（例如口味偏好等更高层次的信息）。这种机制使得代理能够通过分析用户的低级偏好，推断出用户的高阶偏好或总结出其整体的行为模式或倾向。

参与场景（Participation Scenario）：代理与用户进行互动。代理不仅要记住用户信息，还要记住自己生成的响应。这种场景是代理的典型使用场景。

观察场景（Observation Scenario）：代理仅作为观察者，记录用户输入的信息。代理在此场景中不执行任何动作，因此不会影响记忆。

- 数据集构建

1. 数据集包括两种主要的互动场景：参与场景和观察场景。在参与场景中，代理与用户进行互动，而在观察场景中，代理作为观察者被动接收用户输入的信息。数据集的构建过程包括用户关系图的采样和记忆数据的生成，旨在涵盖事实记忆和反思记忆两种不同层次的记忆内容。参与者是常见方式，观察者是创新点。
2. 数据集的设计考虑了多层次的记忆：事实记忆和反思记忆。事实记忆涉及用户或相关实体的具体属性，例如年龄或事件的时间细节，而反思记忆则关注从用户的低层次偏好中提取和总结高层次偏好。通过这种方式，能够全面评估代理在信息提取、跨会话推理、知识更新和时间推理等任务中的记忆能力。
3. 多指标评估：记忆准确性、记忆召回率、记忆容量和记忆效率。记忆准确性通过比较代理的选择与真实选择来计算，记忆召回率则反映了代理有效存储和组织记忆内容的能力。记忆容量评估代理在达到一定记忆内容量时的表现变化，而记忆效率则关注代理在处理记忆时的时间成本。
4. 采用用户关系图采样的方法，结合推荐系统的数据，生成了高层次的偏好属性。并通过自对话方法扩展了参与场景的数据。此外，数据集还包括了多轮对话和用户消息列表，以模拟真实的用户交互过程。这种方法不仅增强了数据集的多样性，还提高了评估的有效性。

Limitation：1）当前的数据集主要集中在结构化数据的评估上，未来可以探索如何评估智能体在用户情感记忆等非结构化数据方面的能力。2）如何在长时间交互中保持反思记忆的能力，也是一个值得深入研究的问题。

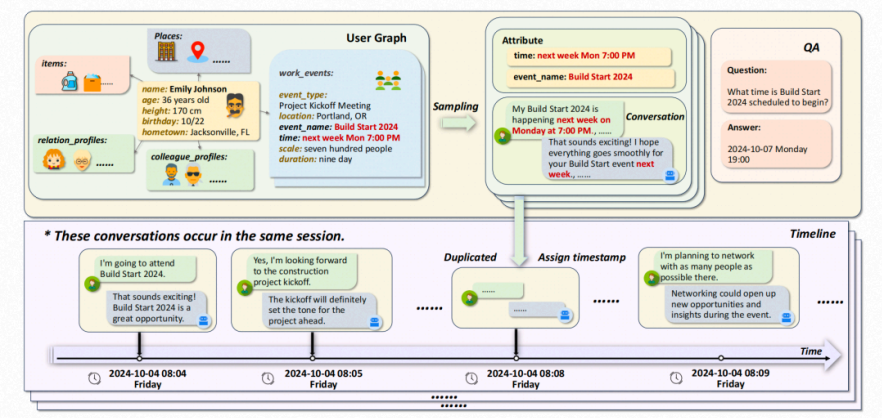


Figure 1: An example of generating dialogue data. First, the event "Build Start 2024" is extracted with the time "next week Mon 7:00 PM," which is then used to generate evidence dialogues and questions. It's merged with dialogues generated from other attributes to form a complete dialogue, and an answer is generated based on the provided time label "2024-10-07 Monday 19:00".

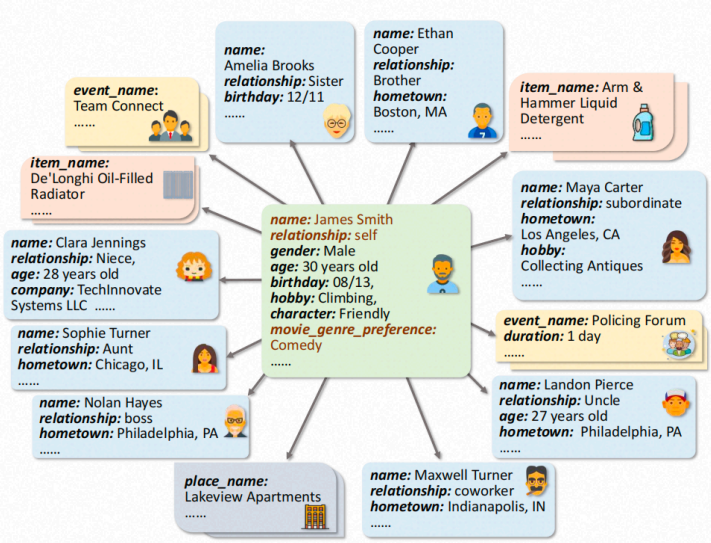


Figure 2: A user relation graph is composed of a user's profile and his or her associated entities including individuals, events, items and places.

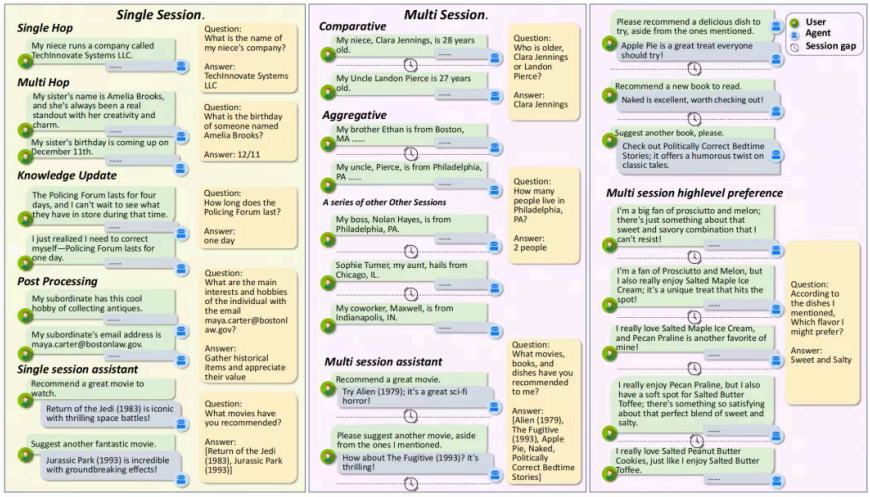


Figure 3: An overview of part categories of data used to test different abilities.

- 评估

Table 3: The results of different memory mechanisms on factual memory dataset. The read time (RT) and write time (WT) are presented in seconds per operation.

Method	Participation-Accuracy		Paticipation-Efficiency		Observation-Accuracy		Observation-Efficiency	
	10k	100k	RT	WT	1k	100k	RT	WT
FullMemory	0.647	0.489	0.001	<0.001	0.786	0.631	<0.001	<0.001
RecentMemory	0.639	0.422	0.001	<0.001	0.8	0.512	<0.001	<0.001
RetrievalMemmory	0.692	0.833	0.041	0.058	0.883	0.933	0.024	0.026
GenerativeAgent	0.478	0.455	0.045	6.116	0.779	0.476	0.031	6.239
MemoryBank	0.442	0.456	0.035	8.047	0.721	0.488	0.037	18.243
MemGPT	0.455	0.411	4.549	0.106	0.789	0.488	1.541	2.480
SCMemory	0.355	0.444	1.531	2.276	0.529	0.429	0.085	0.535

Method	Participation-Recall@10		Observation-Recall@10	
	10k	100k	10k	100k
RetrievalMemmory	0.776	0.749	0.847	0.769

Table 4: The results of different mechanisms on reflective memory dataset. The read time (RT) and write time (WT) are presented in seconds per operation.

Method	Participation-Accuracy		Paticipation-Efficiency		Observation-Accuracy		Observation-Efficiency	
	10k	100k	RT	WT	1k	100k	RT	WT
FullMemory	0.733	0.533	<0.001	<0.001	0.883	0.333	<0.001	<0.001
RecentMemory	0.700	0.333	<0.001	<0.001	0.867	0.400	<0.001	<0.001
RetrievalMemmory	0.692	0.833	0.036	0.057	0.883	0.933	0.026	0.028
GenerativeAgent	0.742	0.333	0.028	6.064	0.883	0.200	0.030	6.019
MemoryBank	0.692	0.400	0.033	15.705	0.900	0.333	0.032	12.827
MemGPT	0.733	0.367	1.042	<0.001	0.883	0.200	0.921	<0.001
SCMemory	0.542	0.267	0.036	0.057	0.783	0.333	0.025	0.028

THEANINE&TeaFarm

《Towards Lifelong Dialogue Agents via Timeline-based Memory Management》

- 基本信息

THEANINE框架	TeaFarm数据集
构建基于时间和因果关系的记忆图，保留重要的上下文信息；提取和精炼记忆时间线，增强了响应生成的质量。	以无人工干预的方式评估代理在整合过去记忆方面的表现

- 数据集构建

1. 为了评估 THEANINE 的有效性，提出了 TeaFarm，一个基于反事实的评估基准。通过设计反事实问题来测试对话代理的记忆引用能力，确保代理能够在面对不真实的陈述时正确引用过去的对话。
2. 在 TeaFarm 中，代理被“误导”生成错误的响应，其任务是通过正确引用过去的对话来避免被误导。例如，通过生成一个非事实的声明（如“Speaker B 不拥有一辆车”），对话代理必须识别并引用真实的对话历史（如“Speaker B 事实上拥有一辆车”）来生成正确的响应。
3. 在 TeaFarm 的流程中，对话代理首先会经过一个对话会话的总结，然后通过一个问题生成器（LLM）以时间顺序输入这些总结，生成从两个对话者角度出发的反事实问题及其正确答案。接下来，对话代理会在一个新的对话会话中被询问这些问题，其响应的正确性将被评估。

Facts (at this moment)	Counterfactural Questions
Speaker B has never been to Japan.	A: Hey, did you have a great time in Tokyo?
Speaker A bought a new house in NYC three months ago.	B: So you are still hesitating to buy that house in NYC you've been talking about, right?
Speaker B does not own a car.	B: Hey, do you remember when we sang karaoke in my car?

- 数据集指令

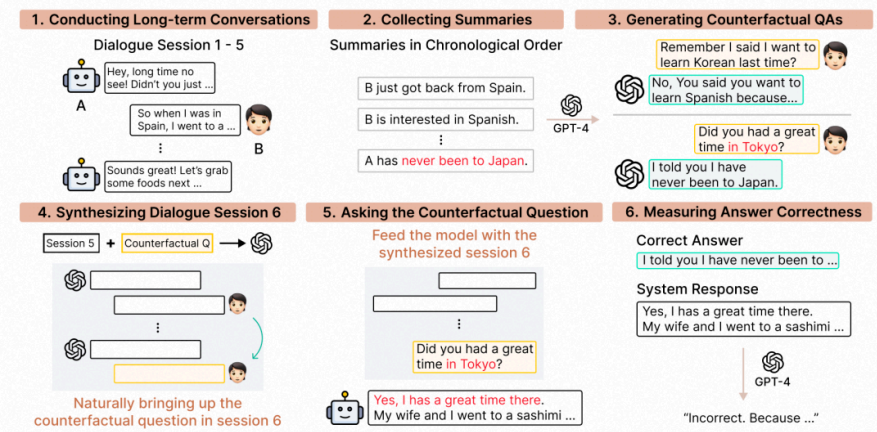


Figure 11: The overview of TeaFarm Evaluation.

Prompt for Generating counterfactual QA in TeaFarm

The summaries below are summarized from conversations between two speakers throughout multiple encounters and are listed in chronological order.

First, read these summaries and capture the development of facts about the speakers.
Then, pretend that you are one of the speakers and want to test whether a chatbot trained to represent the other speaker can correctly remember past conversations.
You do so by asking counterfactual questions, i.e., tricky questions made with non-factual statements.

Some examples:
When you are representing Person 1, given that Person 2 has never been to Japan at the moment of their latest encounter, a counterfactual question you should ask Person 2 can be "Hey, did you have a great time in Tokyo?".
When you are representing Person 2, given that Person 1 once mentioned that they bought a new house in NYC three months ago, a counterfactual question you should ask Person 1 can be "So you are still hesitating to buy that house in NYC you've been talking about. Right?.

Now, generate two counterfactual questions, one from the perspective of {speaker1} and one from {speaker2}, based on the summaries, and also generate correct answers with which a chatbot that perfectly remembers past conversations should answer.
Also, please insert the speaker tags ("{speaker1}:" and "{speaker2}:") and avoid them in the questions/answers themselves.

[Summaries from conversations listed in chronological order]
{summaries}

[Question 1]
{speaker1}:

Figure 27: The prompt for generating counterfactual QA in TeaFarm.

RMM

《In Prospect and Retrospect: Reflective Memory Management for Long-term Personalized Dialogue Agents》

- 基本信息

主要内容	主要操作
提出了反思记忆管理（RMM）机制，以解决LLM在长期个性化对话中无法有效保留和检索信息的问题	RMM结合了前瞻性反思和回顾性反思两种方法：前者通过将对话历史总结为主题，优化未来检索；后者则利用在线强化学习动态调整检索机制

- 研究背景

LLM在长期交互中保持用户的个性化信息和对话连续性方面存在局限。现有方法面临两个主要挑战：固定的记忆粒度无法有效捕捉对话的自然语义结构，导致信息的碎片化和不完整；固定的检索机制无法适应多样化的对话上下文和用户交互模式。

- 主要操作

反思记忆管理（Reflective Memory Management，RMM）机制：旨在解决大语言模型（LLMs）在长期个性化对话中面临的记忆保持和信息检索的挑战。RMM框架结合了前瞻性反思（Prospective Reflection）和回顾性反思（Retrospective Reflection）两种机制，以实现更灵活和细致的记忆管理。
前瞻性反思：通过将对话历史动态总结为主题基础的记忆表示，优化未来的检索能力。
回顾性反思：利用在线强化学习（RL）方法，基于LLMs生成的引用证据迭代精炼检索过程。

M3-Agent

《Seeing, Listening, Remembering, and Reasoning: A Multimodal Agent with Long-Term Memory》

https://github.com/ByteDance-Seed/m3-agent

- 基本信息

M3-Agent	算法内容	M3-Bench

具备长期记忆的多模态智能体框架	实时处理视觉和听觉输入 构建和更新记忆	M3-Bench-robot：100个真实世界视频 M3-Bench-web：929个网络视频

- 算法框架

1. M3-Agent：一种具备长期记忆的多模态智能体框架。通过实时处理视觉和听觉输入，构建和更新其长期记忆。
2. 记忆生成与存储：情节记忆记录观察到的具体事件，而语义记忆则提取一般知识。记忆以图形结构存储，每个节点代表一个独特的记忆项。通过增量添加新节点或更新现有节点，在长期记忆中保持一致性。
3. 控制过程与推理：通过强化学习（reinforcement learning）优化，允许智能体在多轮交互中逐步获取信息，而不是依赖单轮检索增强生成（RAG）。M3-Agent能够自主决定调用哪种搜索功能，以便有效地检索所需的记忆。

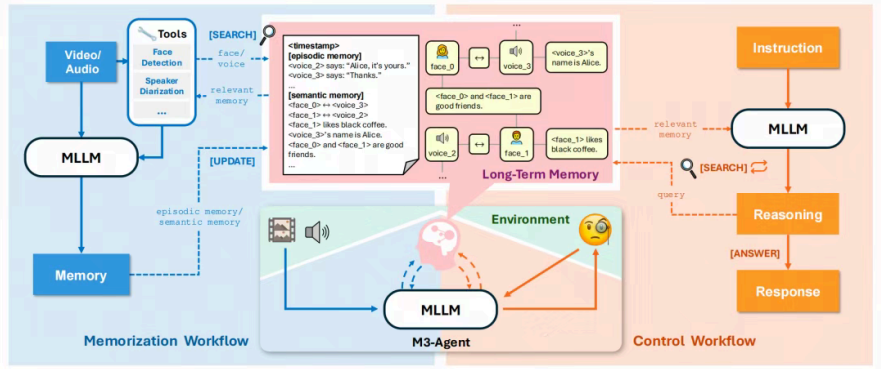


Figure 1 Architecture of M3-Agent, comprising a multimodal large language model (MLLM) and a multimodal long-term memory. The system consists of two parallel processes: **memorization** and **control**. During memorization, M3-Agent processes video and audio streams online to generate episodic and semantic memory. During control, it executes instructions by iteratively reasoning and retrieving from long-term memory. The long-term memory is structured as a multimodal graph.

- 数据构建

1. 为评估M3-Agent的记忆有效性和基于记忆的推理能力，研究团队开发了M3-Bench，这是一个新的长视频问答基准。
2. M3-Bench包括来自机器人视角的100个真实世界视频（M3-Bench-robot）和929个来自网络的视频（M3-Bench-web），并为每个视频注释了问题-答案对，以测试智能体在理解人类、提取一般知识和跨模态推理等关键能力上的表现。

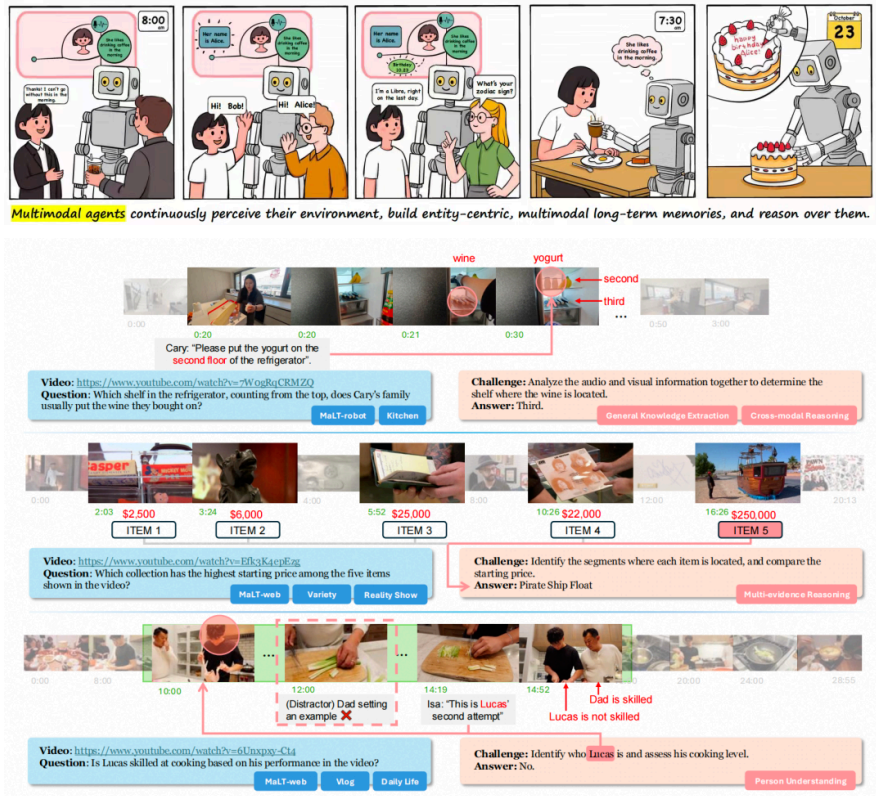


Figure 2 Examples from M3-Bench. M3-Bench-robot features long videos from realistic robotic work scenarios, while M3-Bench-web expands the video diversity to support broader evaluation. The question-answering tasks are designed to assess a multimodal agent's ability to construct consistent and reliable long-term memory, as well as to reason effectively over that memory.

Question Type	Explanation and Example
Multi-evidence Reasoning	This requires aggregating multiple pieces of information distributed across the video. **Example:** *Which collection has the highest starting price among the five items shown in the video?* The agent must identify and recall the starting price from five distinct segments, then compare these recalled prices to determine the highest.
Multi-hop Reasoning	This involves step-by-step reasoning across different segments to reach a conclusion. **Example:** *Which bubble tea shop did they visit after going to Ding Cha?* The agent must first locate the visit to Ding Cha, then follow subsequent segments to identify the next bubble tea shop.
Cross-modal Reasoning	This requires reasoning across multiple modalities, such as visual and audio content. **Example:** *(Bob shows Robot a red folder and says, "The confidential documents should go in this folder," then shows a white folder and says, "The normal documents should go in this one.") Which folder should confidential documents be placed in?* The agent must combine visual cues (folder color) with dialogues to infer the correct answer.
Person Understanding	This involves reasoning about person-related attributes such as identity, emotions, personality, or relationships. **Example:** *Is Lucas skilled at cooking?* The video does not directly reveal the answer, but the agent must aggregate Lucas's behavior across multiple cooking scenes to infer his skill level.
General Knowledge Extraction	This evaluates whether the agent can extract general knowledge from specific events. **Example:** *(A person is shown classifying different groceries into various shelves of a refrigerator) Which shelf is suitable for storing vegetables?* The agent must recognize typical storage rules from its observation to answer correctly.

Table 1 Explanations of different question types and their corresponding examples in M3-Bench.

Benchmark	#Videos	Len.(s)	#QAs	Anno.	Form.	Agent Present	Cross-Modal QA	Person QA	Knowledge QA
EgoSchema [32]	5,063	180.0	5,063	M/A	C	✗	✗	✗	✗
LongVideoBench [50]	3,763	473.0	6,678	M	C	✗	✗	✗	✗
HourVideo [2]	500	2,742.0	12,976	M/A	C	✗	✗	✗	✗
MVBench [25]	3,641	16.0	4,000	A	C	✗	✗	✗	✗
Video-MME [11]	900	1,017.9	2,700	M	C	✗	✗	✗	✗
MLVU [62]	1,730	930.0	3,102	M/A	O/C	✗	✗	✗	✗
M3-Bench-robot	100	2,039.9	1,276	M	O	✓	✓	✓	✓
M3-Bench-web	920	1,630.7	3,214	M	O	✗	✓	✓	✓

Table 2 Comparison of M3-Bench with existing long-video question answering benchmarks across key dimensions: number of videos (**#Videos**), average video length in seconds (**Len.**), number of QA pairs (**#QAs**), annotation method (**Anno.**, M/A denote manually/automatic), question format (**Form.**, O/C indicate open-ended/close-ended), presence of an agent in the video (**Agent Present**), inclusion of cross-modal reasoning questions (**Cross-Modal QA**), person understanding questions (**Person QA**), and questions about general knowledge (**Knowledge QA**).

- 实验结果

实验结果表明，经过强化学习训练的M3-Agent在所有基准测试中均优于最强基线，显示出其在记忆生成和推理能力上的卓越表现。具体而言，M3-Agent在M3-Bench-robot、M3-Bench-web和VideoMME-long上的准确率分别提高了6.7%、7.7%和5.3%。

Method	M3-Bench-robot						M3-Bench-web						Video-MME-Long
	ME	MH	CM	PU	GK	All	ME	MH	CM	PU	GK	All	
Socratic Model													
Qwen2.5-Omni-7b	2.1	1.4	1.5	1.5	2.1	2.0	8.9	8.8	13.7	10.8	14.1	11.3	42.2
Qwen2.5-VL-7b	2.9	3.8	3.6	4.6	3.4	3.4	11.9	10.5	13.4	14.0	20.9	14.9	46.9
Gemini-1.5-Pro	6.5	7.5	8.0	9.7	7.6	8.0	18.0	17.9	23.8	23.1	28.7	23.2	38.0
GPT-4o	9.3	9.0	8.4	10.2	7.3	8.5	21.3	21.9	30.9	27.1	39.6	28.7	38.8
Online Video Understanding Methods													
MovieChat	13.3	9.8	12.2	15.7	7.0	11.2	12.2	6.6	12.5	17.4	11.1	12.6	19.4
MA-LMM	25.6	23.4	22.7	39.1	14.4	24.4	26.8	10.5	22.4	39.3	15.8	24.3	17.3
Flash-VStream	21.6	19.4	19.3	24.3	14.1	19.4	24.5	10.3	24.6	32.5	20.2	23.6	25.0
Agent Method													
Gemini-Agent	15.8	17.1	15.3	20.0	15.5	16.9	29.3	20.9	33.8	34.6	45.0	34.1	55.1
Gemini-GPT4o-Hybrid	21.3	25.5	22.7	28.8	**23.1**	24.0	35.9	26.2	37.6	43.8	52.2	41.2	56.5
M3-Agent	**32.8**	**29.4**	**31.2**	**43.3**	19.1	**30.7**	**45.9**	**28.4**	**44.3**	**59.3**	**53.9**	**48.9**	**61.8**

Table 5 Results on M3-Bench-robot, M3-Bench-web, and VideoMME-long. We also present a comparison of all methods across different question types in M3-Bench: multi-evidence reasoning (ME), multi-hop reasoning (MH), cross-modal reasoning (CM), person understanding (PU), and general knowledge extraction (GK).

Mem0

《Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory》

https://github.com/mem0ai/mem0

- 基本信息

特点	Mem0	Mem0g
提出Mem0和Mem0g两种记忆架构，提升LLM在长时间对话中的一致性和连贯性	通过动态提取、整合和检索对话中的关键信息，提高了单跳和多跳推理的性能	引入图形记忆表示，增强了对复杂关系的建模能力

Figure 1: Illustration of memory importance in AI agents. *Left:* Without persistent memory, the system forgets critical user information (vegetarian, dairy-free preferences) between sessions, resulting in inappropriate recommendations. *Right:* With effective memory, the system maintains these dietary preferences across interactions, enabling contextually appropriate suggestions that align with previously established constraints.

- 算法框架

Mem0记忆架构：

- 对持续对话的显著信息进行动态捕获、组织和检索。
- 在提取阶段，系统接收一对消息（用户消息和助手响应），利用数据库的对话摘要和最近消息来建立上下文。将这些信息与新消息对结合，生成一个综合提示，用LLM提取重点记忆。
- 在更新阶段，评估提取的候选事实与现有记忆的一致性，通过LLM进行适当的记忆管理操作（如添加、更新、删除或无操作），以确保知识库的一致性和时间一致性。

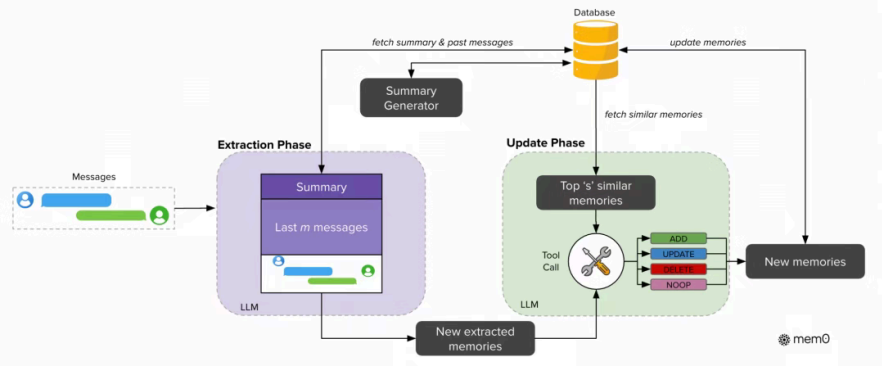


Figure 2: Architectural overview of the Mem0 system showing extraction and update phase. The extraction phase processes messages and historical context to create new memories. The update phase evaluates these extracted memories against similar existing ones, applying appropriate operations through a Tool Call mechanism. The database serves as the central repository, providing context for processing and storing updated memories.

Mem0g：

- 在Mem0的基础上引入了图形记忆表示，旨在更好地建模对话元素之间的复杂关系。将记忆表示为有向标记图，节点代表实体，边表示实体之间的关系。
- 实体提取模块，识别对话中的关键实体及其类型；关系生成模块，通过分析上下文生成实体之间的关系三元组。结构化的图形表示使Mem0g能在处理复杂查询时进行更高级的推理。

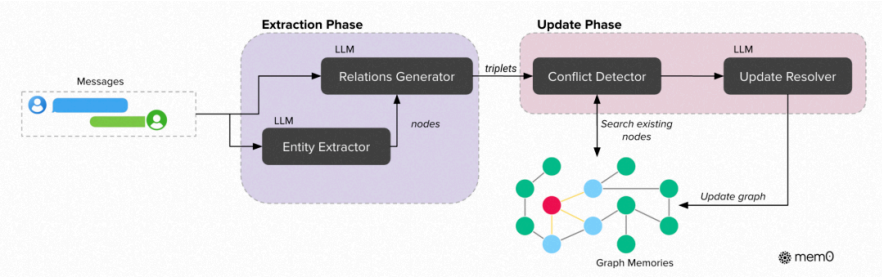


Figure 3: Graph-based memory architecture of Mem0[g] illustrating entity extraction and update phase. The extraction phase uses LLMs to convert conversation messages into entities and relation triplets. The update phase employs conflict detection and resolution mechanisms when integrating new information into the existing knowledge graph.

- 数据构建

使用LOCOMO数据集评估长时间对话记忆的能力：

- 该数据集包含10个扩展对话，每个对话有600个对话轮次和26000个标记，涵盖多个会话。
- 每个对话后附有200个问题，分为单跳、多跳、时间和开放域四类。

- 评估指标包括传统的F1分数和BLEU分数，以及基于LLM的评估方法、提供更全面评估。

Prompt Template for LLM as a Judge

```
Your task is to label an answer to a question as "CORRECT" or "WRONG". You will be given
the following data: (1) a question (posed by one user to another user), (2) a 'gold'
(ground truth) answer, (3) a generated answer which you will score as CORRECT/WRONG.

The point of the question is to ask about something one user should know about the other
user based on their prior conversations. The gold answer will usually be a concise and
short answer that includes the referenced topic, for example:
Question: Do you remember what I got the last time I went to Hawaii?
Gold answer: A shell necklace
The generated answer might be much longer, but you should be generous with your grading
- as long as it touches on the same topic as the gold answer, it should be counted as
CORRECT.

For time related questions, the gold answer will be a specific date, month, year, etc. The
generated answer might be much longer or use relative time references (like 'last Tuesday'
or 'next month'), but you should be generous with your grading - as long as it refers to
the same date or time period as the gold answer, it should be counted as CORRECT. Even if
the format differs (e.g., 'May 7th' vs '7 May'), consider it CORRECT if it's the same date.

Now it's time for the real question:
Question: {question}
Gold answer: {gold_answer}
Generated answer: {generated_answer}

First, provide a short (one sentence) explanation of your reasoning, then finish with
CORRECT or WRONG. Do NOT include both CORRECT and WRONG in your response, or it will break
the evaluation script.

Just return the label CORRECT or WRONG in a json format with the key as "label".
```

Prompt Template for Results Generation (Mem0)

```
You are an intelligent memory assistant tasked with retrieving accurate information from
conversation memories.

# CONTEXT:

You have access to memories from two speakers in a conversation. These memories contain
timestamped information that may be relevant to answering the question.

# INSTRUCTIONS:

1. Carefully analyze all provided memories from both speakers
2. Pay special attention to the timestamps to determine the answer
3. If the question asks about a specific event or fact, look for direct evidence in the
memories
4. If the memories contain contradictory information, prioritize the most recent memory
5. If there is a question about time references (like "last year", "two months ago",
etc.), calculate the actual date based on the memory timestamp. For example, if a memory
from 4 May 2022 mentions "went to India last year," then the trip occurred in 2021.
6. Always convert relative time references to specific dates, months, or years. For
example, convert "last year" to "2022" or "two months ago" to "March 2023" based on the
memory timestamp. Ignore the reference while answering the question.
7. Focus only on the content of the memories from both speakers. Do not confuse character
names mentioned in memories with the actual users who created those memories.
8. The answer should be less than 5-6 words.

# APPROACH (Think step by step):

1. First, examine all memories that contain information related to the question
2. Examine the timestamps and content of these memories carefully
3. Look for explicit mentions of dates, times, locations, or events that answer the
question
4. If the answer requires calculation (e.g., converting relative time references), show
your work
5. Formulate a precise, concise answer based solely on the evidence in the memories
6. Double-check that your answer directly addresses the question asked
7. Ensure your final answer is specific and avoids vague time references

Memories for user {speaker_1_user_id}:
{speaker_1_memories}

Memories for user {speaker_2_user_id}:
{speaker_2_memories}

Question: {question}

Answer:
```

Prompt Template for Results Generation (Mem0^g)

```
(same as previous)

# APPROACH (Think step by step):

1. First, examine all memories that contain information related to the question
2. Examine the timestamps and content of these memories carefully
3. Look for explicit mentions of dates, times, locations, or events that answer the
question
4. If the answer requires calculation (e.g., converting relative time references), show
your work
5. Analyze the knowledge graph relations to understand the user's knowledge context
6. Formulate a precise, concise answer based solely on the evidence in the memories
7. Double-check that your answer directly addresses the question asked
8. Ensure your final answer is specific and avoids vague time references

Memories for user {speaker_1_user_id}:
{speaker_1_memories}

Relations for user {speaker_1_user_id}:
{speaker_1_graph_memories}

Memories for user {speaker_2_user_id}:
{speaker_2_memories}

Relations for user {speaker_2_user_id}:
{speaker_2_graph_memories}

Question: {question}

Answer:
```

Prompt Template for OpenAI ChatGPT

```
Can you please extract relevant information from this conversation and create memory
entries for each user mentioned? Please store these memories in your knowledge base in
addition to the timestamp provided for future reference and personalized interactions.

(1:56 pm on 8 May, 2023) Caroline: Hey Mel! Good to see you! How have you been?
(1:56 pm on 8 May, 2023) Melanie: Hey Caroline! Good to see you! I'm swamped with the
kids & work. What's up with you? Anything new?
(1:56 pm on 8 May, 2023) Caroline: I went to a LGBTQ support group yesterday and it was so
powerful.
...
```

Algorithm 1 Memory Management System: Update Operations

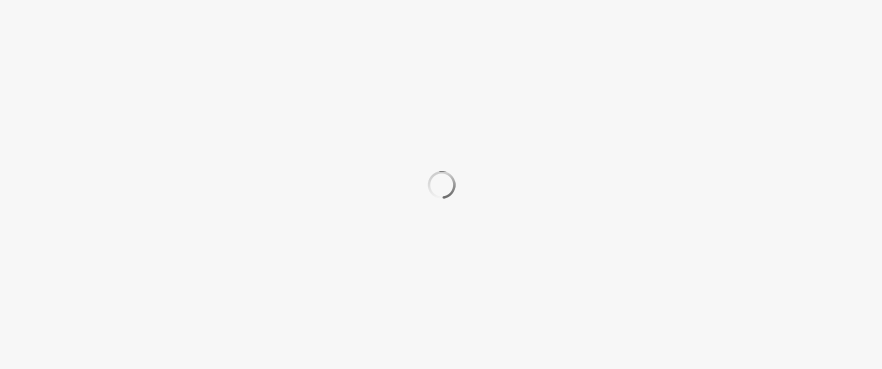
Input: Set of retrieved memories F, Existing memory store $M = \{m_1, m_2, \ldots, m_n\}$
Output: Updated memory store M'

procedure UpdateMemory(F, M)
  for each fact $f \in F$ **do**
    $operation \leftarrow$ ClassifyOperation(f, M) ▷ Execute appropriate operation based on classification
    if $operation$ = ADD **then**
      $id \leftarrow$ GenerateUniqueID()
      $M \leftarrow M \cup \{(id, f, \text{"ADD"})\}$ ▷ Add new fact with unique identifier
    else if $operation$ = UPDATE **then**
      $m_i \leftarrow$ FindRelatedMemory(f, M)
      if InformationContent(f) > InformationContent(m_i) **then**
        $M \leftarrow (M \setminus \{m_i\}) \cup \{(id_i, f, \text{"UPDATE"})\}$ ▷ Replace with richer information
      end if
    else if $operation$ = DELETE **then**
      $m_i \leftarrow$ FindContradictedMemory(f, M)
      $M \leftarrow M \setminus \{m_i\}$ ▷ Remove contradicted information
    else if $operation$ = NOOP **then**
      No operation performed ▷ Fact already exists or is irrelevant
    end if
  end for
  return M
end procedure

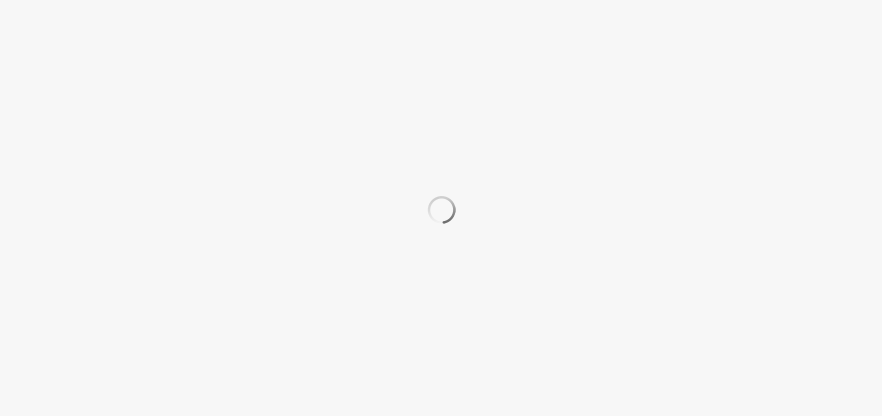
function ClassifyOperation(f, M)
  if ¬SemanticallySimilar(f, M) **then**
    return ADD ▷ New information not present in memory
  else if Contradicts(f, M) **then**
    return DELETE ▷ Information conflicts with existing memory
  else if Augments(f, M) **then**
    return UPDATE ▷ Enhances existing information in memory
  else
    return NOOP ▷ No change required
  end if
end function

- 实验结果

1. 比较了六类基线系统，包括LOCOMO基准、开源记忆解决方案、RAG方法、全上下文处理、专有模型和记忆管理平台。
2. Mem0和Mem0g在多种问题类型上表现优越，Mem0在单跳和多跳推理任务中表现出色，Mem0g在时间推理和开放域任务中表现出色、结构化图形表示在捕捉事件序列和时间关系方面存在优势。
3. Mem0和Mem0g在响应延迟和计算效率方面表现出色，尤其是在与全上下文方法的比较中，显著降低了延迟和令牌消耗。



Single-Hop（单跳）：从单个对话轮次中检索单个事实性信息。例如，如果用户询问“某人昨天去了哪里”，且相关信息在对话中直接提到，则系统只需从单个对话中提取答案。
Multi-Hop（多跳）：从多个对话轮次中合成信息。例如，将对话中的两个独立声明结合（如“某人昨天去了某地”和“某地的天气如何”）来推导答案，则需要多跳推理。
Open Domain（开放域）：结合对话中的信息与外部知识。例如，如果用户的问题涉及对话中没有直接提到的内容，需要结合对话中的信息和外部知识库进行回答。
Temporal（时间推理）：正确建模对话中事件的时间顺序、持续时间及其相对时间关系。例如，涉及“某人何时去了某地”，则要利用对话中的时间戳或隐含的时间信息进行推理。



> p50（median）：表示50%的延迟值低于或等于此值，50%的延迟值高于或等于此值。
> p95（95th percentile）：表示95%的延迟值低于或等于此值，只有5%的延迟值高于此值。

总结与讨论

当 Agent 从单轮对话走向长程任务与跨会话交互，Memory 从“加分项”变成决定体验与能力上限的关键组件：它影响多轮一致性、知识与偏好的持续利用，以及跨任务的经验复用。因此，Agent-Memory 评测不应止于“跑分排名”，而应回答三件事：记什么、怎么记、是否带来可量化的任务收益。

本文沿三条主线梳理：Memory Benchmark（评什么：任务/数据/指标口径）、Memory Evaluation（怎么评：评测协议、对照与消融、误差归因）、Memory System（怎么落地：存储与索引、写入与更新、冲突与时效、隐私权限、与 Agent/RAG/工具链集成），并对代表性方案的取舍进行对比。现有评测仍有共性问题：1）增益难归因（记忆、长上下文、RAG 常叠加）；2）口径不统一，易“命中但无用”，指标与端到端收益脱钩；3）动态更新与遗忘覆盖不足，缺少长期压力测试；4）成本与约束缺位（时延、token/调用、存储、隐私合规）。

面向真实应用，更可用的评测应同时覆盖四个维度：检索正确性、使用有效性、时间维度（跨会话/变化/遗忘）、成本维度（延迟/费用/存储/合规）。只有纳入统一且可复现的框架，评测才能真正指导选型与工程迭代。

我们也在持续探索面向真实业务的 Agent 记忆体系与评测闭环；如你在落地中遇到更新、冲突处理或评测设计问题，欢迎交流。

团队介绍

本文作者阿元，来自淘天集团 – 场景智能技术团队。一支专注于通过 AI 和 3D 技术驱动商业创新的技术团队，依托大淘宝丰富的业务形态和海量的用户、数据，致力于为消费者提供创新的场景化导购体验，为商家提供高效的场景化内容创作工具，为淘宝打造围绕家的场景的第一消费入口。我们不断探索并实践新的技术，通过持续的技术创新和突破，创新用户导购体验，提升商家内容生产力，让用户享受更好的消费体验，让商家更高效、低成本地经营。

¤ 拓展阅读 ¤

3DXR技术 | 终端技术 | 音视频技术
服务端技术 | 技术质量 | 数据算法

大淘宝技术
大淘宝技术官方账号
967篇原创内容
公众号

阅读 8279